Notebook investigations for explorEDA

11 inspected examples / 7 Observable notebooks / 4 computational essays and investigations

The best references build an argument one result at a time: a question leads to a calculation, its output changes the next question, and the reader can inspect the steps. The selected examples cover simple summaries, grouped comparisons, time series, missing-data sensitivity, model evaluation and spatial statistics. Each example begins with captured evidence of the actual page.

Recommended first reproductions

- Small first proof: What's different? Timeseries forecast performance

- Broader representative workflow: Discovering Date Patterns

- Stretch case: Is Seattle Really Seeing an Uptick In Cycling?

Selection and evidence

- Seven of the eleven selections are Observable notebooks. Eleven is deliberate: quality and inspectability took priority over adding a twelfth marginal example.

- Four Observable selections share an electricity-analysis collection but exercise different patterns: subtraction, decomposition, calendar features and joins/conditional comparisons. The two penguin examples likewise show a progression from grouped exploration to model-informed revision.

- The main reading sequence matters in every selection. Some individual outputs contain facets, adjacent maps or a chart/table switch. Reproducing the essay does not require arranging unrelated charts in a dashboard grid.

- No example requires elaborate animation for its analytical core. Maps, resampling and model fitting remain meaningful capability questions, not assumed features of explorEDA.

How to read the evidence

- Actual analysis pages were opened and inspected in their rendered state. Screenshots are captures of those pages, not gallery thumbnails or generated mockups. Some captures show a revealed solution, changed control, or later analytical step; captions identify those states.

- Observable source was inspected through visible cells and, where useful, archived exports of the author's notebook source checked against the live page. An archived revision is identified when used. Search snippets alone were not treated as evidence.

- Observable's visible JavaScript was inspected as evidence of source methods. Equivalent expressions, queries and chart configurations may reproduce those methods; these references do not imply that explorEDA needs a general JavaScript runtime.

- Data availability distinguishes a working live attachment/table, an inspected downloadable snapshot, and an upstream source that was linked but not independently retrieved. Exact formulas are described only when

supported by visible source or documented methods.

- Observed controls are labeled separately from proposed explorEDA controls. Every compatibility statement is a hypothesis based on the supplied capabilities; explorEDA itself was not inspected. Effort ratings are relative scope judgments, not schedules.

- Two retained notebooks have localized execution failures: Bratenahl's sensitivity output and two spatial-tutorial cells. Their functioning analytical sequences and visible methods remain useful; the failures and reproduction limits are specified in their entries.

Screened out

- The Olympics age-by-sport companion notebook currently displayed errors instead of its plots, so it was excluded.

- An AWS EC2 pricing investigation had a missing CSV attachment and failed chart outputs. It was excluded despite a promising narrative.

- A Taylor Swift Plot workshop had working visual controls but limited investigative argument and failing SQL outputs. It was screened out of the main set.

- Data releases, single-chart examples and generic platform demos were not added merely to fill the target count.

Reading order

1. Plot Exploration: Penguins

2. What's different? Analyzing Time-Series Forecast Performance

3. What caused this? An analysis adventure

4. Discovering Date Patterns

5. Correlation over time

6. How Bratenahl, Ohio, Disproportionately Targets Black Drivers

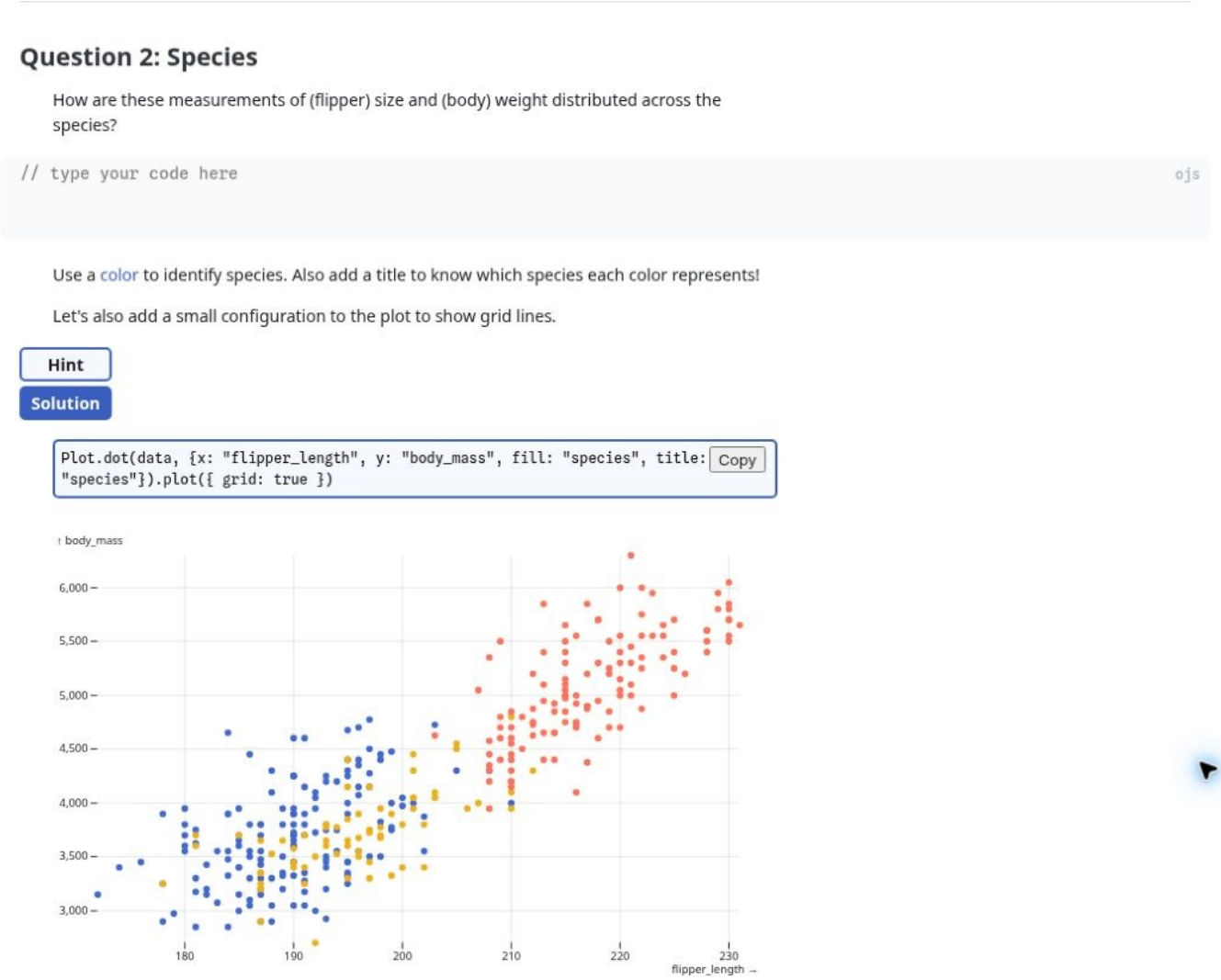
7. Local Spatial Autocorrelation Interactive Tutorial

8. Is Seattle Really Seeing an Uptick In Cycling?

9. Get started with tidymodels and #TidyTuesday Palmer penguins

10. Many countries have decoupled economic growth from CO_2 emissions, even if we take offshored production into account

11. Film Dialogue from 2,000 screenplays, Broken Down by Gender and Age



01. Plot Exploration: Penguins - Screenshot 1. Question 2, with its solution expanded: the prose motivates species as a grouping variable, editable calculation/configuration text precedes the scatterplot, and the next question follows vertically. Direct source. Browser side margins cropped; page content is unchanged.

Observable / Raw observations > grouped comparison > faceted distributions > summary overlay

What the analysis does

The notebook investigates how penguin body measurements vary by species, sex and island. A raw mass-versus-flipper relationship becomes more informative when species color and then sex facets are added; later questions compare counts, weight distributions and medians. In the inspected charts, longer flippers tend to accompany greater mass, and the groups separate differently across the comparisons. That is our visual interpretation. The author supplies questions and executable solutions rather than a concluding biological claim. This is a guided investigation using real research data, so it is strongest as a small, progressive reproduction, with the pedagogical character stated explicitly.

Role of prose. Each question gives the preceding result a reason to change: first relate two measurements, then distinguish species without discarding the relationship, then add sex without losing the species encoding. Later prose motivates counting, binning and median overlays. Hints explain the calculation or encoding; interpretation is largely left to the reader. The notebook therefore demonstrates a sequence of analytical decisions more strongly than a finished computational essay.

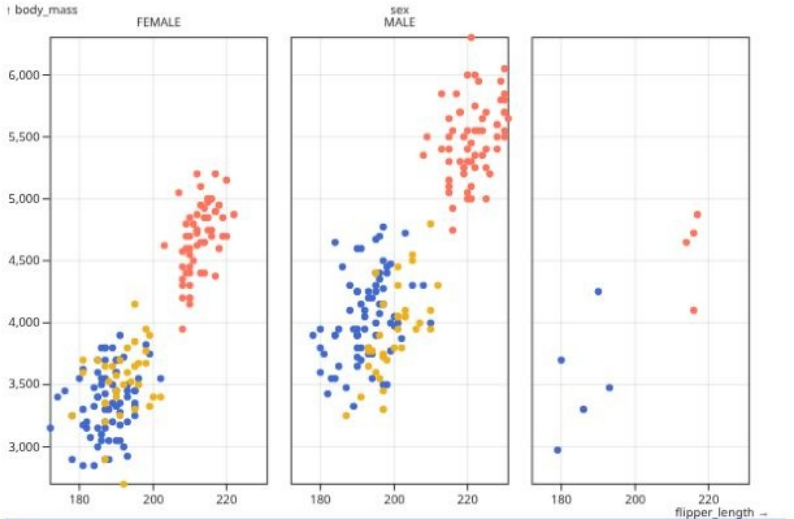
We could use a color to identify sex, like in the previous chart. But we can also reuse the previous code, and add a facet on sex.

Hint

Solutions

```
Plot.dot(data, {x: "flipper_length", y: "body_mass", fill: "species", title: "species"}).plot({
  facet: {
    data,
    x: "sex"
  },
  grid: true,
  marks: [
    // you can add more marks to the plot with the marks array
    Plot.frame(),
  ]
```

Copy




```
Plot.plot({
  facet: {
    data,
    x: "sex"
  },
  grid: true,
```

Copy

01. Plot Exploration: Penguins - Screenshot 2. The next analytical step retains species color while adding sex facets. The visible definition and complete output share the reading column. The unlabeled third facet contains observations with missing sex. Direct source. Browser side margins cropped; page content is unchanged.

Source data and calculations

Original source. Kristen Gorman and Palmer Station Long Term Ecological Research; the Palmer Penguins dataset distributed by Allison Horst and colleagues, with a notebook-specific CSV attachment.

Row grain and scope. One measured penguin; 344 records spanning Adelie, Chinstrap and Gentoo penguins on Biscoe, Dream and Torgersen islands in the Palmer Archipelago, Antarctica. Canonical provenance covers 2007-2009; the notebook attachment omits year.

Fields and units. species, island, bill_length, bill_depth, flipper_length, body_mass and sex. Lengths are millimeters; mass is grams. The attachment uses MALE/FEMALE and blank sex values, and removes unit suffixes from column names. The canonical CSV retains year, _mm/_g suffixes and lowercase sex values.

- Exact attached palmer-penguins.csv
- Original dataset documentation and attribution
- Canonical CSV source file
- Canonical CSV direct download
- Original study

Calculations, dependencies and stated limits

- The attachment loads with typed CSV parsing. No explicit row-cleaning step precedes the inspected comparisons. Missing measurements and sex need an explicit policy in a reproduction; the sex-faceted solution visibly retains a blank category.

- Questions 1-3 use raw observations: x = flipper_length, y = body_mass; Question 2 adds fill and hover title = species; Question 3 keeps those encodings and facets horizontally by sex.

- Question 4 groups by species and counts records for a bar chart. Question 5 repeats species counts within island facets; an alternative uses x = island, y = species and radius = count.

- Question 6 bins body_mass and counts rows using Plot.binX with thresholds: 20. Questions 7-8 facet these histograms by sex horizontally and species vertically. The threshold is a requested bin parameter; do not assume it guarantees exactly twenty rendered bins under the library's defaults.

- Question 8 overlays a full-height median tick on each sex/species distribution using Plot.groupZ({x: 'median'}, {x: 'body_mass'}), within the existing facet context.

- Question 9 asks how many females exceed the lightest male. Its supplied solution shows male extrema, female median and individual observations by species. It does not produce the requested count, so an exact numerical answer would be an extension rather than an observed output.

What feeds later results. The shared data table feeds every chart. The analytical dependency is progressive - each comparison motivates the next - but the original chart cells generally repeat their definitions rather than consuming the preceding chart's output. Within the statistics chart, the histogram counts and median reducer share the same raw rows and facet groups.

Reproduction sufficiency. The actual attachment link, loaded table and visible solution definitions are sufficient for the core reproduction. Data provenance is public and documented. The canonical CSV is an accessible substitute if the attachment fails: rename columns, normalize sex values and preserve missingness. This changes schema preparation, not the intended sample, but identity with the attachment should be checked before claiming a byte-for-byte match.

Limits and uncertainty. This is a teaching investigation, with fewer authored conclusions than the journalism examples. Native table sorting and row selection are visible, but no source dependency links the table selection to the charts. Hint/Solution toggles change presentation; they are not analytical filters. The exact Question 9 count is absent.

Elements needed to reproduce it

1. Introduce the measured penguins > load and inspect rows > establish fields and missing values.

2. Ask whether heavier birds have longer flippers > configure a raw scatter > inspect the association.

3. Ask whether species explain the pattern > add species color and hover labels > compare clusters.

4. Ask about sex while retaining species > facet the same scatter by sex > compare within groups.

5. Ask how sample composition varies > count species, then species within islands > inspect bars or a count table.

6. Ask about weight distributions > bin mass by sex/species > overlay group medians > inspect overlap and formulate the next question.

Element	Inputs	Calculation or configuration	Visible result	Interaction	Dependency
Essential: question and transition text	Study question; result of prior comparison	Explain why one additional variable or statistic is needed	Text before and after each analytical block	None required	Sequential reasoning, without forcing a code dependency

Element	Inputs	Calculation or configuration	Visible result	Interaction	Dependency
Essential: data and preview	Attached seven-field CSV	Typed parsing; explicit missing-value policy; field units	Inspectable table	Sorting; selection need not filter charts	Shared source for all comparisons
Essential: progressive scatter	flipper_length, body_mass, species	x = flipper length mm; y = mass g; fill/title = species; grid	Full-width scatter; species identification	Hover title observed; visual encoding edits proposed	Raw data; changes to this definition remain local
Essential for grouped comparison: sex facets	Same rows and scatter definition; sex	Horizontal facet = sex; retain species colors and common axes	Female, male and missing-sex panels within one chart block	Facet configuration edits proposed	Same data and field policy as prior scatter
Essential for broader reproduction: counts	species, island	Group by species; count rows; optionally facet by island	Species bars and island comparisons; consistent color legend	Grouping/category order edits proposed	Raw data > grouped counts
Essential for statistical extension: histogram and median	body_mass, sex, species	Count within mass bins, thresholds = 20; horizontal sex and vertical species facets; median tick overlay	Faceted distributions with reference marks	Bin, facet and overlay edits proposed	Counts and medians must share grouping and filters
Optional: teaching controls and alternatives	Question text; solution definitions	Reveal hints/solutions; alternative bubble and stacked views	Progressive disclosure and alternate encodings	Hint/Solution buttons observed	Presentation state only

Fit for explorEDA

Smallest faithful version. Reproduce Questions 1-3 with the data table: a mass/flipper scatter, a second block adding species color, and a third adding sex facets, joined by the original analytical questions. This is a small first proof. Counts and a histogram-plus-median can extend it when grouped calculations are ready.

Likely fit, subject to verification. Tables, interactive charts, filters, calculated fields and saved state appear sufficient for much of the exploration, based only on the product description. No arbitrary JavaScript execution is implied by these requirements.

Capabilities to investigate. Facets with shared scales, grouped counts and medians, histogram bin defaults, overlay marks, explicit missing-category behavior, and whether copied or derived chart definitions preserve common configuration without coupling every later edit.

Graphical edits that should update text. Visual changes to x/y fields, species color, sex or island facet, category order, opacity, hover fields, axis units, bin thresholds and median overlays should update the defining text. A switch from raw observations to counts must update the calculation as well as the chart encoding. The text should expose the grouping and denominator, not merely record cosmetic changes.

Control scope. The original exposes table inspection and solution toggles, not a shared analytical filter. A proposed global species filter would affect every dependent result; changing one chart's color or facet should remain local unless the text explicitly shares that setting. Histogram and median layers need identical group/filter scope.

Execution and editing lesson. This tests whether users can express a familiar calculation/visual combination in text, refine it with graphical controls, and read a meaningful sequence afterward. It also tests the distinction between a shared dataset, a copied chart definition and a true dependent calculation. Missing values must remain legible when a visual edit introduces a facet.

"medium" and "small".

Let's take this a step further by changing what we're measuring on the y-axis, to focus even more on the differences themselves.

Previously the y-axis represented the actual or forecasted energy demand. In the chart below the y-axis represents the *difference* in energy usage between the actuals and forecast.

So, if the forecast was accurate, the value would be 0. If the **actuals were above the forecast** by 5000 kW, we would see an orange mark from 0 to 5000. And, if the **actuals came in below the forecast** by 1000 kW, we'd see a purple mark from 0 to -1000.



Admittedly in the difference chart we do lose some information. We no longer can see if the actual demand was relatively high or low. On the other hand, we're able to more clearly compare the differences over time because they share a common baseline. Moreover, smaller differences that were harder to see are now noticeable.

Did you notice?

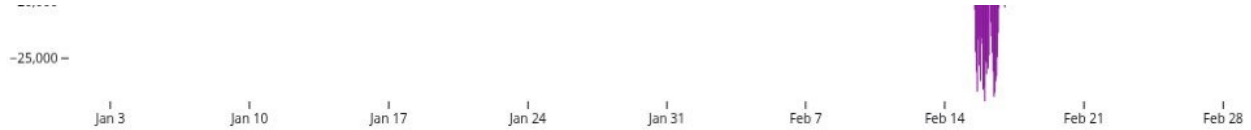
02. What's different? Analyzing Time-Series Forecast Performance - Screenshot 1. The analytical turning point: prose defines the signed difference between actuals and forecast, the chart plots that calculated quantity against a zero baseline, and the next paragraph interprets the result. Direct source. Browser side margins cropped; page content is unchanged.

Observable / Actual versus forecast > signed difference > event-window investigation

What the analysis does

The notebook investigates hourly Texas demand around the February 2021 power crisis. Subtracting the forecast from actual demand makes two dips before the rolling blackouts noticeable. This is a particularly clear example of an analytical argument produced by changing what is calculated and displayed. It does not establish why those early dips occurred; price is offered as a follow-up hypothesis.

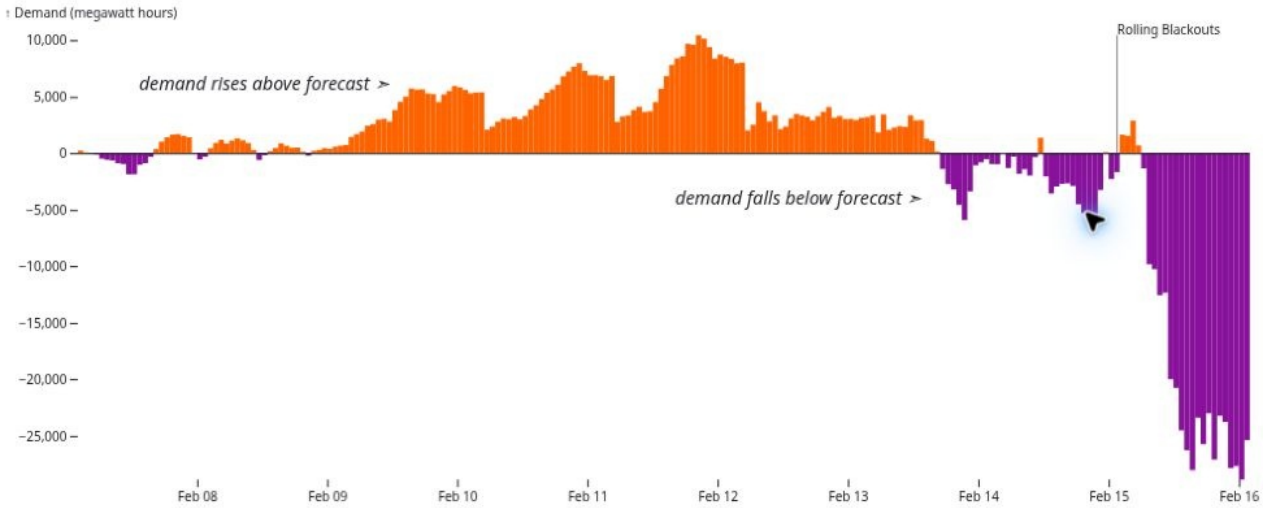
Role of prose. The prose introduces an apparent comparison problem, asks whether demand was above or below forecast, explains the signed difference, interprets the pre-event pattern, and motivates a closer date window. The later diagrams explain why comparing two sloping lines can obscure that difference.



Admittedly in the difference chart we do lose some information. We no longer can see if the actual demand was relatively high or low. On the other hand, we're able to more clearly compare the differences over time because they share a common baseline. Moreover, smaller differences that were harder to see are now noticeable.

Did you notice?

One thing that jumped out at me is that the demand actually started to fall below the forecast *just prior* to the blackouts, as we can see in this "zoomed in" view focused on just the week prior. Moreover, there were two "dips" during the 36 hours before the blackouts were initiated.



Now that I know what to look for, I can see it when I go back to the original chart. But, sometimes **it's not enough for an important feature of the data to be visible. It needs to be noticeable.**

02. What's different? Analyzing Time-Series Forecast Performance - Screenshot 2. Later in the same reading sequence, the residual calculation is reused in an event-window zoom; the prose interprets two pre-blackout dips. Direct source. Browser side margins cropped; page content is unchanged.

Source data and calculations

Original source. U.S. Energy Information Administration Open Data, an ERCOT/Texas hourly demand and day-ahead-forecast snapshot downloaded in August 2021 and attached in September 2021.

Row grain and scope. One hourly observation in Texas/ERCOT, January-February 2021; the live raw and renamed arrays contain 1,417 rows. The fixed event-window subset contains 216 rows.

Fields and units. date (UTC timestamp), demand renamed actuals, and forecast. The chart labels demand in MWh, but some explanatory text uses kW; retain raw values and reconcile this source labeling inconsistency rather than applying an invented conversion.

- Exact attached Texas demand/forecast CSV

- EIA original provider

- Author's EIA download notebook

Calculations, dependencies and stated limits

- Parse the attached CSV with typed fields; rename demand to actuals.

- Compute signed difference actuals - forecast; positive and negative values receive consistent orange and purple encodings.

- Represent rolling blackouts at 2021-02-15T07:25:00Z (1:25 a.m. CST). Filter the zoom from 2021-02-07T07:25:00Z through 2021-02-16T07:25:00Z inclusively.

- The source derives daily/weekly ticks from UTC intervals shifted six hours to show Central time, and derives maxima for chart/annotation placement (maxDelta 10,436 and maxValue 76,783 in the inspected state).

What feeds later results. Typed data > renamed relation > actual/forecast line charts and residual charts. The zoom predicate produces a smaller relation reused by later line and residual comparisons. Shared event and semantic-color definitions feed multiple graphics and explanatory annotations.

Reproduction sufficiency. The live loaded rows, downloadable attachment link, visible transformation definitions and stated subtraction are enough for the analytical core. The attachment bytes were not independently downloaded. A failed browser download did not prevent the notebook itself from loading and displaying the data.

Limits and uncertainty. Unit labels are inconsistent; do not preserve them uncritically. The event window is a fixed authored filter, not a reader-operated brush. The early demand decline is an observed temporal relationship, not a demonstrated cause. Missing-value and duplicate-hour handling were not independently audited; an exact match should make those policies explicit.

Elements needed to reproduce it

1. Introduce the forecast question > inspect hourly actual/forecast values > show two lines and blackout annotation.

2. Explain the desired comparison > derive actuals minus forecast > show signed differences around zero.

3. Interpret the unexpected pre-blackout dips > filter to the nine-day event window > show the enlarged residual pattern > propose a follow-up question.

Element	Inputs	Calculation or configuration	Visible result	Interaction	Dependency
Essential: question and interpretation text	Forecast question and chart evidence	Explain signed error and connect each result to the next question	Prose before and after each output	None observed	Interpretations refer to residual and zoom results
Essential: source and residual calculation	date, demand, forecast	Typed load; rename demand; difference=actuals-forecast	Inspectable original/derived records	Table inspection if reproduced	Shared relation feeds all views
Essential: original comparison	Hourly date, actuals, forecast; event	x=time; y=demand; two labeled colored lines; vertical event rule	Full-column time series	No reader control observed	Shared data, event and colors
Essential: signed-difference plot	Date and signed difference	x=time; y=difference; zero baseline; orange above, purple below	Above/below-forecast pattern	No reader control observed	Depends on residual calculation
Essential: event-window result	Shared relation and start/end timestamps	Inclusive date filter; same difference encoding; callouts for pre-event dips	Magnified nine-day comparison	Fixed authored window; an editable date control would be an addition	Derived subset feeds later residual and line comparisons
Optional: perception diagrams and filled bands	Original series and difference concept	Fill between lines; authored explanatory illustrations	Why a common baseline helps	None	Can be omitted without losing the investigation

Fit for explorEDA

Smallest faithful version. Keep the introduction, original two-line chart, explicit residual calculation, signed-error plot and nine-day zoom with interpretation. The conceptual cartoons and repeated return to the original chart are optional.

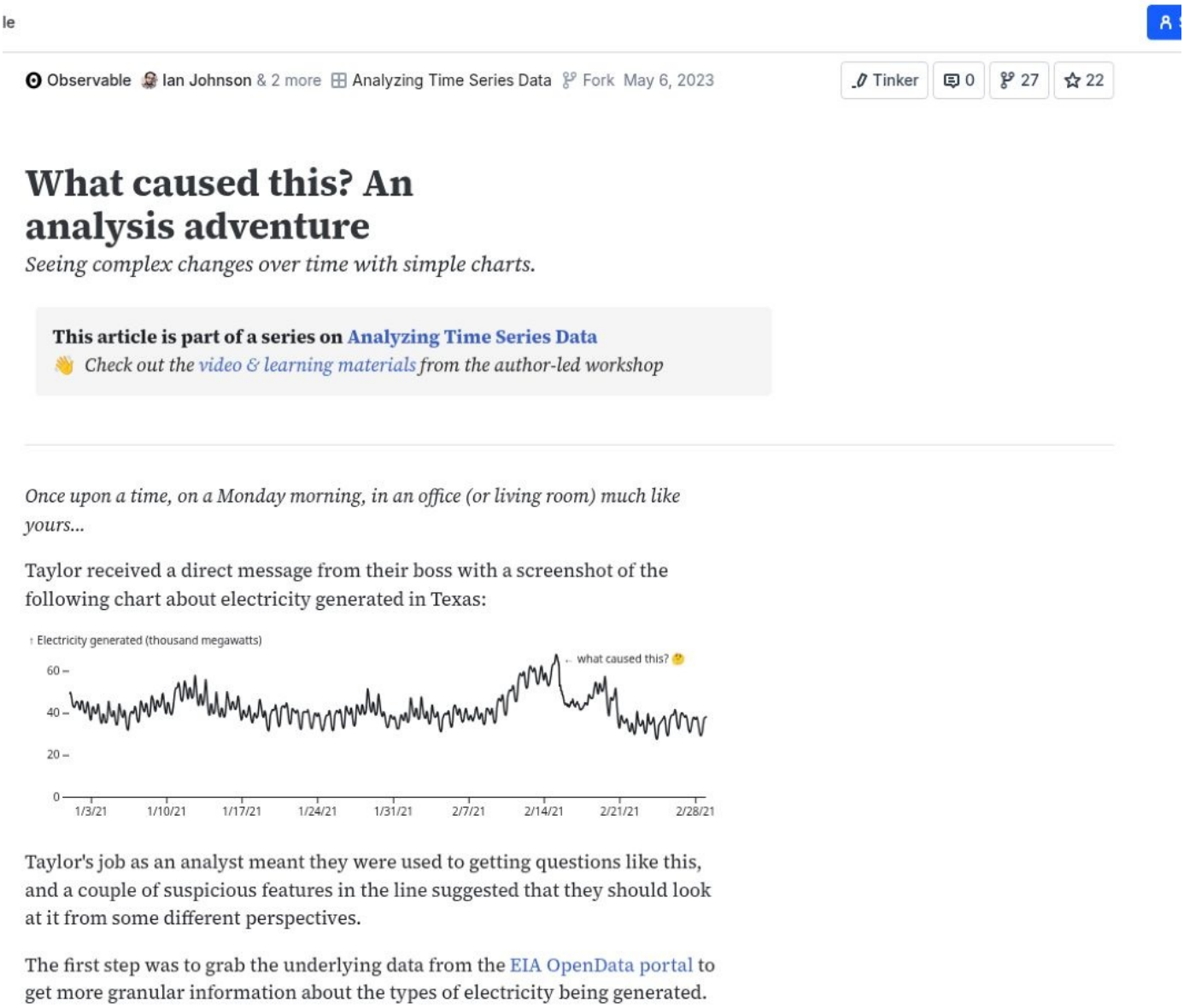
Likely fit, subject to verification. Calculated fields, filters, interactive chart definitions, tables and saved analysis state appear to cover the core, based only on the product description.

Capabilities to investigate. Shared named calculations across blocks; positive/negative rule or bar encodings; event and text annotations; a clear distinction between display-domain edits and upstream date filtering; consistent units and semantic colors.

Graphical edits that should update text. Graphical changes to mark type, axis fields and limits, color assignments, event position/label, annotation text and window bounds should update the relevant block text. Changing a residual chart's visual style should preserve the subtraction it depends on.

Control scope. There are no main reader controls in the inspected notebook. The authored event filter affects the zoomed branch, while the complete series remains available above.

Execution and editing lesson. A small calculation can be the analytical turning point. This tests whether two-way editing preserves a named derived field, reuses it in a later block, and records local view changes without accidentally changing the evidence above.



03. What caused this? An analysis adventure - Screenshot 1. The opening anomaly: a concrete question, the total-generation chart, and prose proposing a more granular data source. Direct source. Browser side margins cropped; page content is unchanged.

Observable / Aggregate anomaly > category decomposition > shared-scale comparisons

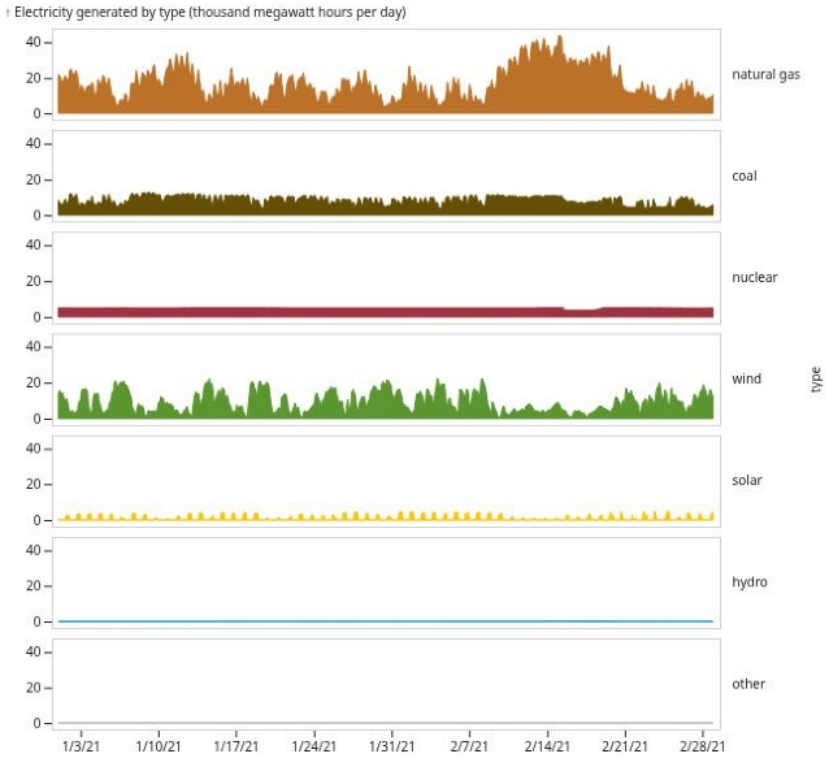
What the analysis does

A fictional analyst investigates a spike in real Texas generation data. Successive views show a two-week episode, not just one high day: a coal plateau, a nuclear dip, sustained lower wind generation, and a large rise in natural-gas generation. The epilogue explicitly says the aggregate patterns cannot determine which fuel is intrinsically better or worse or by themselves establish the crisis mechanism.

Role of prose. The story gives each chart a reason to exist: confirm the total, identify contributing categories, address overlapping lines, examine stacking, and finally separate fuel types with comparable scales. Interpretation then accompanies each selected fuel panel.

this was going to make any sense to other people, they would need to split out each series into it's own graph.

In the past, Taylor had struggled with creating what are sometimes referred to as "small multiples", it was laborious and time consuming to recreate the same chart over and over with small differences. On top of that there were easy mistakes to make like forgetting to maintain the same y-axis domain and creating misleading comparisons. Luckily, Taylor had just read about faceting, an option in Plot that does the breaking out of sub-plots in just a couple lines of code.



This was it! By considering each fuel type on it's own, but in the context of the others Taylor would be able to identify what patterns were happening, as well as how relevant they were to the overall situation.

03. What caused this? An analysis adventure - Screenshot 2. The later decomposition: prose explains the need for common scales, followed by fuel-type facets using the same hourly records. Direct source. Browser side margins cropped; page content is unchanged.

Source data and calculations

Original source. EIA Open Data hourly net generation by fuel for the Texas region; a cached API response deliberately attached so the article remains reproducible.

Row grain and scope. One fuel category × UTC hour in January-February 2021, Texas region. The live transformed array has 9,912 rows, covering seven fuels: coal, natural gas, nuclear, other, solar, hydro and wind.

Fields and units. series_id/id, series name/name, derived type, parsed UTC date, and value. The y-axis divides values by 1,000. Source labels vary between thousand megawatts and thousand megawatt hours per day; the facet code uses hourly rows directly, so a daily-total interpretation is unsupported.

- Exact attached texasCrisis2021.json

- EIA original provider

- EIA grid monitor used for fuel colors

Calculations, dependencies and stated limits

- Load cached JSON; flatten each series' date/value pairs; extract fuel type from the series name; parse %Y%m%dT%HZ as UTC; sort by date.

- The total-generation line bins by UTC hour and sums value across fuel records. Its opening variant applies a window with k=1, which does not smooth the data.

- Later fuel lines and stacked areas map the same date/value rows with type determining grouping and color. Faceting splits by type while retaining a common y scale.

- Selected-fuel panels highlight the February 7-21 episode. Exact styling/shading helper logic was not fully inspected; the underlying fuel comparison and scale transforms were visible.

What feeds later results. One cleaned long relation feeds the total aggregation, fuel lines, stacked areas, facets and individual fuel views. A shared semantic fuel-color mapping also supports colored phrases in the prose.

Reproduction sufficiency. The cached 215KB JSON link, populated table, visible load/cleaning source, and inspected chart definitions support the core reproduction. Bytes were not independently downloaded. Reproducing the original obsolete EIA API URL is unnecessary because the frozen attachment is present.

Limits and uncertainty. The source's hourly-versus-daily and power-versus-energy wording should be resolved explicitly. The notebook decomposes measured generation; it is not a causal model of generation failures, unmet demand or fuel reliability. Missing-value and duplicate-hour handling were not independently audited; an exact match should make those policies explicit.

Elements needed to reproduce it

1. Receive an anomaly question > acquire detailed hourly fuel data > inspect the records.

2. Sum fuel categories per hour > reproduce the total line > ask what contributes.

3. Split lines by fuel > try stacked area > explain why overlap and moving baselines hinder comparison.

4. Facet with common scales > inspect selected fuels in separate full-width panels > interpret the two-week episode and state causal limits.

Element	Inputs	Calculation or configuration	Visible result	Interaction	Dependency
Essential: narrative question and transitions	Anomaly and successive chart findings	Explain why each representation prompts the next	Prose above and below each chart	None observed	Forms the analytical sequence
Essential: cleaned long relation	Cached series JSON	Flatten; extract fuel; parse UTC; sort date	Fuel-hour table	Ordinary table inspection	Shared source for every output
Essential: total-generation calculation/line	Date, fuel, value	Group/bin UTC hour; sum value; x=date; y=total/1,000	Single total line with anomaly label	None observed	Depends on cleaned relation
Essential: category comparison	Date, type, value; shared color map	x=date; y=value/1,000; color/group=type	Overlaid fuel lines	No reader controls observed	Same rows, new configuration
Essential: separated fuel result	Same rows and categories	Facet rows by type; shared y-domain; area or line marks; ordered fuel labels	Comparable fuel-specific patterns	None observed	Preserves common data and scale semantics
Essential: interpretation panels	Selected coal/nuclear/wind/gas rows	One selected series per reading block; highlight February episode	Specific findings attached to each fuel	Fixed authored selections	Selection is local to each result
Optional: stacked-area intermediate	Same rows	Stack y by fuel and color by type	Composition view explaining why facets help	None	Useful transition; can be omitted in a compact proof

Fit for explorEDA

Smallest faithful version. Keep the anomaly, source table, hourly total, category lines and fuel facets, followed by the four short findings. An initial reproduction can omit the fictional office framing, stack intermediate and custom highlight styling.

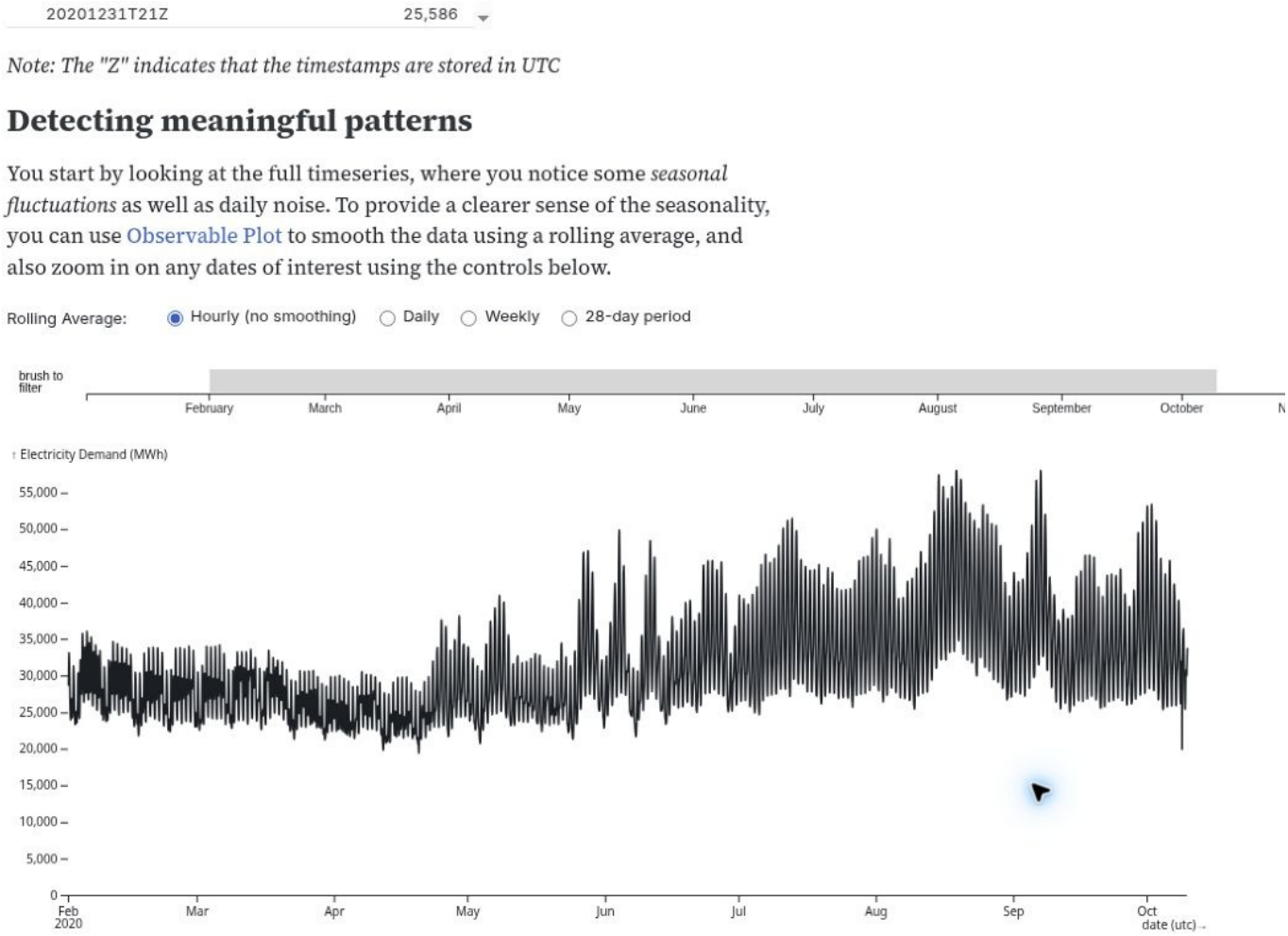
Likely fit, subject to verification. Grouping, aggregation, charts, calculated fields and filters appear compatible with the described capabilities; this remains a hypothesis.

Capabilities to investigate. Common-scale faceting, stable category color mappings shared with prose, named aggregate versus detail relations, and preserving independently authored comparisons while editing chart configurations.

Graphical edits that should update text. Switching total to grouped lines, line to area, stacked to faceted layout, fuel order, scale domain, color or annotation should persist into the block text. Removing grouping is a calculation/configuration change with analytical consequences and should remain explicit.

Control scope. No analytical reader controls were observed. All panels reuse one dataset with different authored aggregations, grouping or local fuel selections.

Execution and editing lesson. This tests whether visually editing a chart can progressively refine an investigation while keeping prior steps readable and preserving aggregation grain, shared scales and categories.



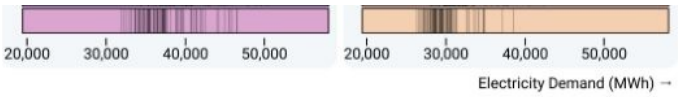
04. Discovering Date Patterns - Screenshot 1. The initial investigation: explanatory prose, a rolling-average selector, a date brush and a full-width demand chart. Captured with hourly data selected. Direct source. Browser side margins cropped; page content is unchanged.

Observable / Time series > calendar features > grouped distributions

What the analysis does

The question is how much structure can be discovered from only a timestamp and a measured value. The notebook first smooths California electricity demand, then derives local calendar categories and compares distributions within them. Its reported patterns vary by hour and season; the inspected quarter-by-day view also makes weekday/weekend differences visible. This is an analytical walkthrough with interpretation, rather than a set of unrelated charts.

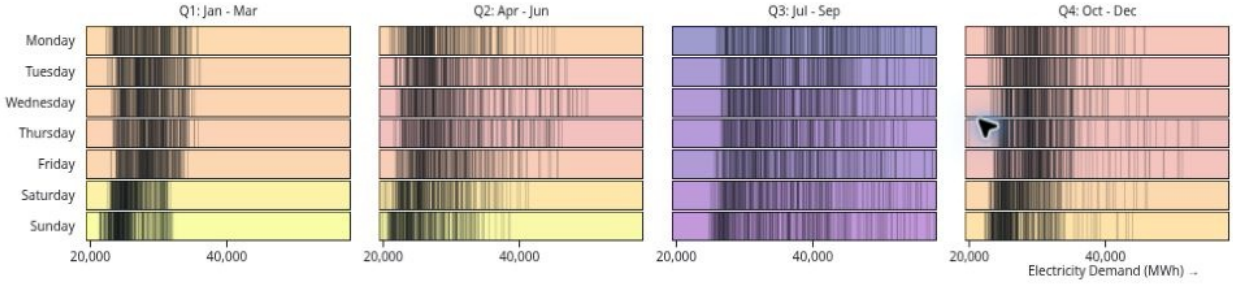
Role of prose. The prose introduces the restricted starting information, explains why smoothing helps, motivates additional date fields, and uses the first results to choose the next comparison. It also distinguishes UTC display time from local calendar time and closes by explaining how the same reasoning transfers to other timestamp/value data.



Using Observable, you create an interactive visualization so others can explore the data by selecting *Hour*, *Day*, *Weekend*, *Week*, *Month*, *Quarter*, or *None* to define the columns and rows of the chart (also known as *faceting*).

Break out Horizontally by: ○ Hour ○ Day ○ Weekend ○ Week ○ Month ◉ Quarter ○ None

Break out Vertically by: ○ Hour ◉ Day ○ Weekend ○ Week ○ Month ○ Quarter ○ None




```
▸ Object {date: 2021-01-01T07:00Z, hour: 23, day: "Thursday", weekend: "Weekday", week: 53, month: "December", quarter:
data[0]                                                                                                    ojs
```

💡 *If you try "Day" and "Week", you'll see every day of the year!*

Closing

Even when your data is "just" a timestamp and a value, you can explore it deeply through rolling averages with different windows or by augmenting the data into familiar time-based categories like quarters or weekends. Different breakdowns will reveal different aspects of the data.

Want to do see this analysis with your own data? Just replace the `raw_data` cell with your own data in the same format (two columns: `date`, and `value`).

More in the Analyzing Time Series Data collection:

04. Discovering Date Patterns - Screenshot 2. A later dependent step, captured after changing the vertical grouping to Day. The horizontal grouping remains Quarter; prose, controls, the result and a derived-record inspection stay in one reading sequence. Direct source. Browser side margins cropped; page content is unchanged.

Source data and calculations

Original source. U.S. Energy Information Administration Open Data; an attached California-region hourly-demand snapshot used by the notebook.

Row grain and scope. One California-region hourly demand observation; 8,784 rows for the local 2020 leap year. The final local-hour record is stored as 2021-01-01T07:00Z.

Fields and units. date (UTC timestamp) and value (electricity demand, chart-labelled MWh). Derived fields are local hour, day of month, weekday, weekend/weekday, ISO week, month and quarter, using America/Los_Angeles.

- Exact attached california.csv
- EIA original provider
- Author's EIA download notebook
- Archived authored source, revision 1760

Calculations, dependencies and stated limits

- Parse the timestamp and derive calendar features in America/Los_Angeles, preserving the original instant for the time axis.

- The rolling transform uses k = 1, 24, 168 or 672 hourly observations. The source delegates alignment and edge handling to its Observable Plot version; those defaults were not independently resolved.

- The date brush supplies the line's x-domain and masks points outside the interval. It does not filter the data used by the later grouped output.

- The later chart groups raw observations by the two selected calendar categories. Each group displays demand values as ticks and its mean as the colored background; group means also feed tooltips. Weekday and month orders are explicit.

What feeds later results. raw_data > enriched data > inspected table and grouped distributions. enriched data + smoothing choice + brush limits > line chart. Grouping selectors affect the faceted result and tooltip summaries, while the earlier smoothing and brush controls remain local to the line.

Reproduction sufficiency. The live table, attached CSV link and visible/archived authored transformations are sufficient for the core reproduction. The attachment link was inspected and the notebook loaded its rows; the CSV bytes were not independently downloaded. The archive is a preserved revision and was checked against live behavior, not assumed to be the current page wholesale.

Limits and uncertainty. Resolve window alignment, edge behavior and sort order explicitly for an exact match. The source uses a PST tooltip label throughout the year despite deriving daylight-saving-aware local fields; a faithful analytical reproduction should label the timezone clearly. This is a labeling issue, not evidence of incorrect local feature derivation. Missing-value and duplicate-hour handling were not independently audited; an exact match should make those policies explicit.

Elements needed to reproduce it

1. Explain the timestamp-only question > load and inspect date/value rows.

2. Define a rolling window and visible date extent > show the time series > interpret daily and seasonal structure.

3. Explain local calendar features > derive them > display the enriched table.

4. Choose two grouping dimensions > show demand distributions and group means > compare seasonal and weekday patterns.

Element	Inputs	Calculation or configuration	Visible result	Interaction	Dependency
Essential: question and method text	Analysis question; field definitions	Explain UTC versus local time and the purpose of each comparison	Narrative above and below outputs	None observed	Interpretation refers to the relevant result
Essential: source and feature calculation	Hourly date, value	Parse date; derive America/Los_Angeles calendar fields	Inspectable table with original and derived fields	Table inspection	Feeds line and grouped output
Essential: rolling line and controls	Enriched rows; k; date limits	x=UTC time; y=demand MWh; k-row rolling mean; stable y extent	Full-width line plus axis note	Hourly/daily/weekly/28-day radios; date brush	Controls affect this output only
Essential: grouped distribution	Enriched rows; horizontal and vertical categories	x=demand; fx/fy=chosen calendar groups; ticks=rows; fill=group mean	Calendar comparison with ordered facets and mean tooltip	Two grouping selectors; hover	Recomputes grouping and means, independent of line controls
Essential: interpretation and next question	Observed distributions	Explain how the chosen categories expose another pattern	Reading transition and conclusion	None observed	Must remain associated with its comparison

Element	Inputs	Calculation or configuration	Visible result	Interaction	Dependency
Optional: annotated example images	Selected chart states	Callouts illustrating hour and quarter differences	Worked interpretations between major steps	None	Can be replaced by ordinary text and annotations

Fit for explorEDA

Smallest faithful version. Use the two-column snapshot, a raw-data table, one rolling line and one quarter-by-weekday grouped chart. Keep the text that motivates the move from time to calendar categories. A mean-by-group chart can be an initial simplification, but it should be labeled as losing the distribution information.

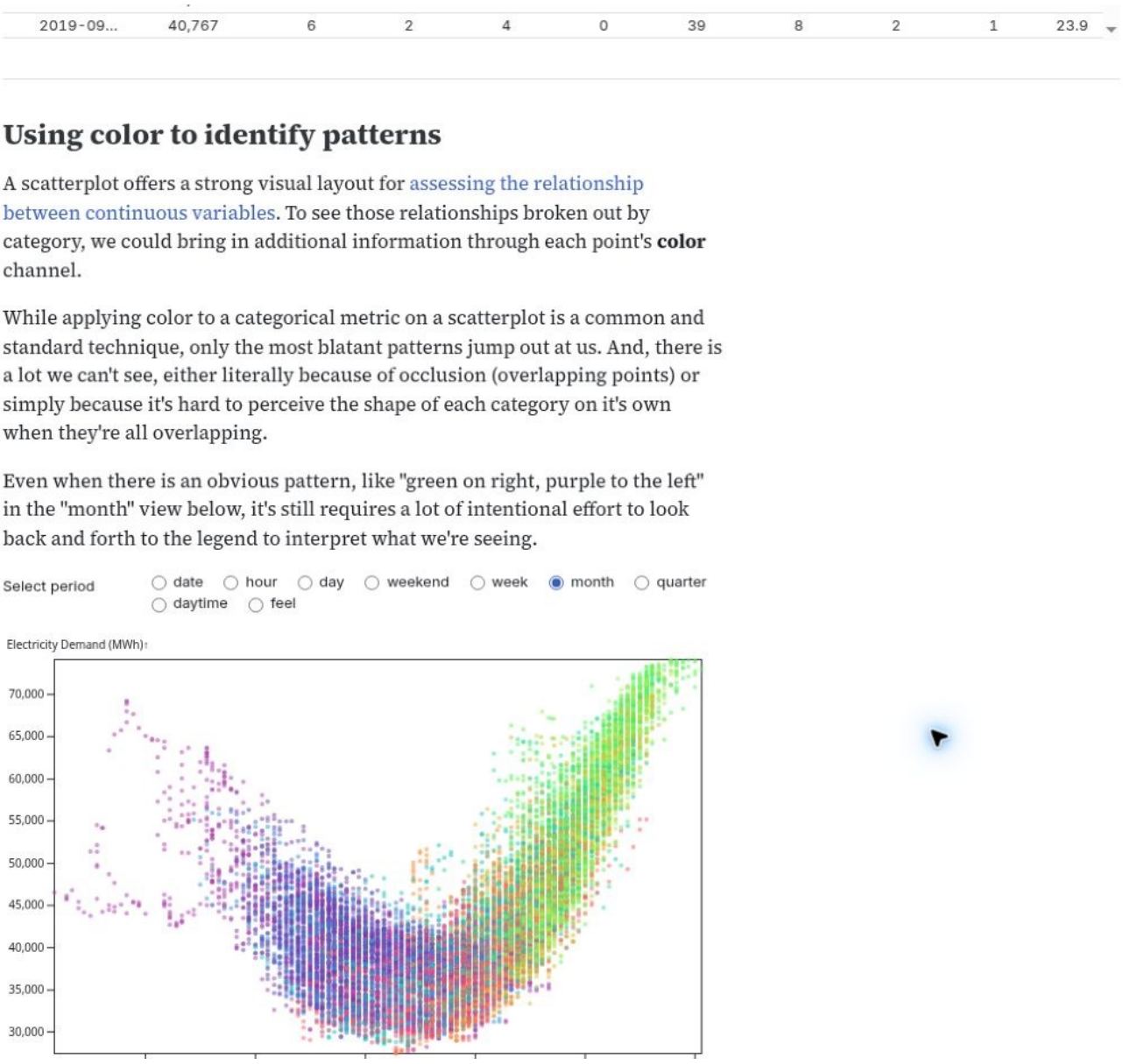
Likely fit, subject to verification. Tables, calculated calendar fields, filters, interactive charts and saved state appear to cover much of this workflow, based only on the supplied product description.

Capabilities to investigate. Timezone-aware date extraction, rolling windows with explicit defaults, two-dimensional facets, per-group distribution marks, and control bindings to a specific result. These are capability questions rather than requirements for arbitrary JavaScript execution.

Graphical edits that should update text. Changing the smoothing window, axis domain, facet fields, category order, tick opacity, mean color scale, labels or tooltip content should update the corresponding block definition. A facet change that changes aggregation should update both grouping and result configuration, rather than only its visual layout.

Control scope. Observed controls have different scopes: smoothing and brush are local to the line; the later selectors recompute only grouped comparisons. A future global date filter would be a different, explicitly declared behavior.

Execution and editing lesson. The reading sequence and calculation dependencies are related but not identical. An enriched table can feed separate branches, each with local control state. Two-way editing should preserve those branches and expose defaults such as window size and calendar timezone in text.



05. Correlation over time - Screenshot 1. Prose motivates a month-colored scatterplot and exposes the limitations of overlapping groups; the table from the preceding calculation remains visible at the top. Direct source. Browser side margins cropped; page content is unchanged.

Observable / Time-aligned join > visual relationship > calendar-conditioned comparisons

What the analysis does

The notebook asks how the relationship between Texas electricity demand and temperature changes over time. It moves from a crowded colored scatterplot to calendar facets. Its worked interpretation compares 10 a.m. with other hours: demand is relatively higher at low temperatures and lower at high temperatures. The analysis is visual; the current notebook does not calculate a linear correlation coefficient or regression.

Role of prose. The prose defines the comparison, explains the join's geographical assumption, diagnoses occlusion and legend effort, motivates spatial separation, and interprets the shapes against the full-data background.

To address this we might try different color schemes, hoping to find one that feels more insightful, better fits the data, or meets our color perception needs. Using the selector below, you can swap out different colorschemes.

Color Scheme: Cyclical, Nameable, High Perceptual Differentiation (rainbow)

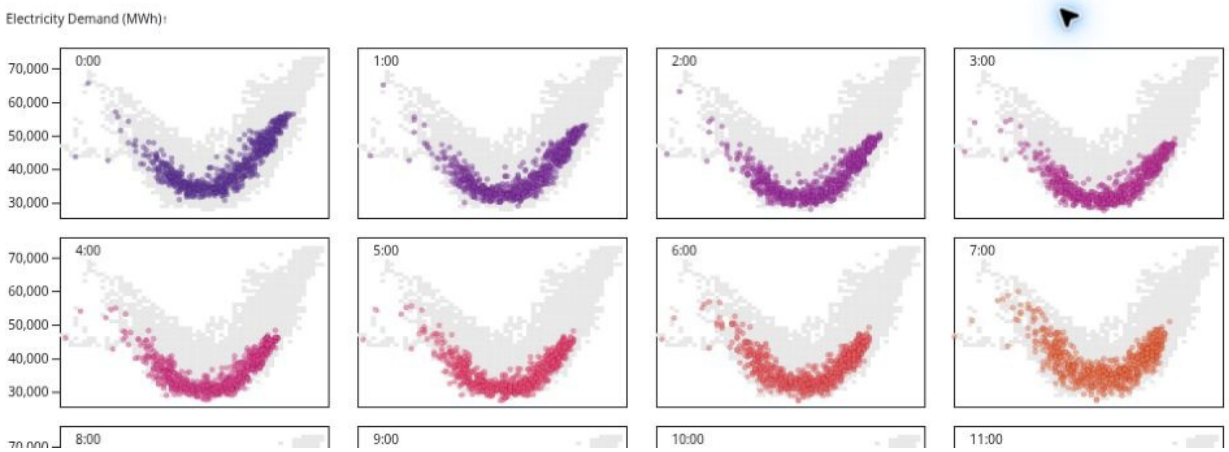
The default is a cyclical colorscheme, **rainbow**, that uses many hues and therefore has high perceptual differentiation for people who don't experience color vision deficiency. The colors are also easily "name-able", as we recognize them as "green", "purple", "red", etc, which makes them easier to talk about with collaborators or stake-holders. Click through to see other options, including '**magma**' which is an attractive palette ranging from dark to light, '**cividis**' which is optimized for those with limited color perception, or the diverging palette **'RdYlBu'** that runs from red through yellow to blue.

While you might find a palette you like better, it doesn't solve the crux of the issue. It's still hard to see the forest for the trees (or, more literally, the 2-distributions for the dots).

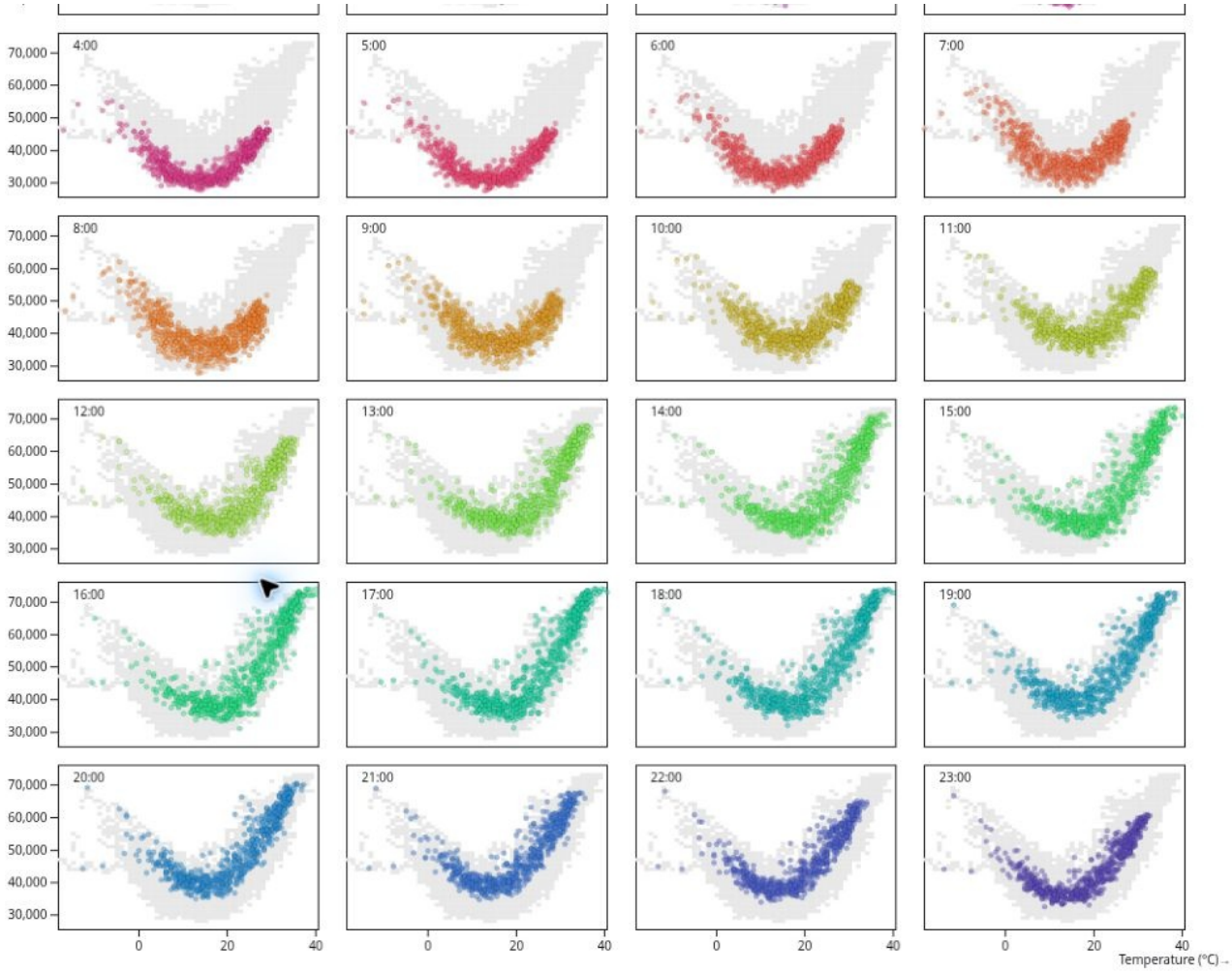
Using facets to identify patterns

As an alternative to using color, we can use Observable Plot to break the data out into different groups — also known as *facets* — to expose the relationships.

Select period ○ month ◉ hour ○ quarter ○ weekend ○ daytime ○ feel



05. Correlation over time - Screenshot 2. A later step after selecting hour: explanation, the control and the beginning of the faceted result. The same overall distribution appears as gray reference marks in every facet. Direct source. Browser side margins cropped; page content is unchanged.



The base of each chart is the same: a light grey heatmap showing the temperature and electricity demand range for the whole year.

On top of that base we draw brightly colored dots for that specific facet's data.

This way we can see the shape of the 2-d distribution for each category (each

05. Correlation over time - Screenshot 3. The hourly facets continue into an explanation of how each subset compares with the shared reference distribution. Direct source. Browser side margins cropped; page content is unchanged.

Source data and calculations

Original source. EIA Texas electricity demand and NOAA Dallas/Fort Worth airport weather, downloaded through the authors' companion notebooks and attached as CSVs.

Row grain and scope. One matched UTC hour from late September 2019 to late September 2021. Live demand contains 17,545 rows; the temperature and joined arrays each contain 17,528. The complete archived weather CSV matching the live attachment hash was inspected, confirming the period.

Fields and units. date; value renamed demand (MWh); air_temp renamed temp (°C); precomputed hour, day, weekend, week, month, quarter, daytime and feel. Demand covers Texas while temperature is from one airport.

- Exact attached demand CSV
- Exact attached weather/features CSV
- NOAA original hourly-weather source
- Author's weather download notebook
- Archived authored source, revision 1969

Calculations, dependencies and stated limits

- Typed-load both CSVs; join on their shared UTC hourly timestamp; rename measured fields and drop series id/name.

- Color the scatter by a selected time category. Random drawing order and opacity 0.5 reduce the effect of occlusion.

- The faceted view groups by a separately selected category. A constant-gray binned occupancy layer of the entire dataset is repeated in every facet; source bin widths are 1°C and 1,000 MWh. Colored dots show each subset.

- Calendar category labels and ordering are explicit; a shared palette feeds both the scatter and facets.

What feeds later results. Joined data feeds the table and both chart branches. One radio controls scatter color grouping; another controls faceting. The palette selector is shared. The gray background deliberately references all joined rows rather than the current facet subset. One finer dependency appears in archived revision 1969: the first scatter category also controls whether the Cividis palette is reversed in the faceted view. That styling dependency was found in source, not tested as a live interaction.

Reproduction sufficiency. The actual controls worked and the join, attachments, schema and preserved source are visible. The archive was checked against the live narrative, data hashes and behavior; it is a pinned revision, not presumed identical to every current UI detail.

Limits and uncertainty. The exact cutoffs of precomputed daytime and feel categories were not verified; use the attached columns or restrict a minimum version to explicit hour/month categories. A statewide demand series and one airport temperature series have different geographic scope, an assumption the notebook itself discusses. Missing-value and duplicate-hour handling were not independently audited; an exact match should make those policies explicit. The timezone used to create the attached calendar-category columns was not independently established; use the supplied fields for the reference reproduction. Joined row counts alone do not verify one-to-one matching or identify all unmatched-hour cases.

Elements needed to reproduce it

1. Ask about a changing relationship > load and align hourly sources > display the joined table.

2. Color temperature-versus-demand by a time category > inspect overlap > try alternate palettes.

3. Separate calendar categories into facets > repeat the full-data gray reference > compare hourly distributions and interpret the 10 a.m. pattern.

Element	Inputs	Calculation or configuration	Visible result	Interaction	Dependency
Essential: question and join caveat	Source definitions and analytical question	Explain time alignment and spatial mismatch	Introductory text and footnote	None	Qualifies all later interpretation
Essential: join and inspection	Two UTC-hour CSVs	Inner/shared-field join; rename demand/temp; drop id/name	Joined hourly table	Table inspection	Feeds both comparisons
Essential: scatter and category control	Joined rows; selected period	x=temp °C; y=demand MWh; color=time category; opacity 0.5; radius 2	Categorical relationship plot and matching legend	Radio: date/hour/day/weekend/week/month/quarter/daytime/feel	Local color-grouping branch
Essential: facets and reference	Joined rows; separate facet category	Same x/y; facet=category; gray all-data 1°C×1,000-MWh occupancy bins; colored subset dots	Comparable distributions against a fixed reference	Radio: month/hour/quarter/weekend/daytime/feel	Local facet branch plus full-data reference

Element	Inputs	Calculation or configuration	Visible result	Interaction	Dependency
Essential: palette and labels	Category order and selected scheme	Shared color scale; sequential/categorical legend behavior	Consistent colors and readable labels	Palette dropdown	Updates both charts; archived first-period setting also affects Cividis reversal
Essential: interpretation	Facet shapes and gray reference	Explain relative demand at a given temperature and hour	Prose following the evidence	None	Depends on chos en/illustrated comparison
Optional: illustrated explanation	Selected hour comparisons	Static callouts for 5 a.m., 10 a.m. and 6 p.m.	Worked visual interpretation	None	Can be ordinary annotations instead

Fit for explorEDA

Smallest faithful version. Join the two attached sources, show a month-colored scatter, then an hourly faceted scatter with an all-data reference and the accompanying interpretation. Four selected hours could be a compact first pass if clearly identified as a subset.

Likely fit, subject to verification. Tables, interactive encodings, calculated fields, filters and saved state plausibly cover much of this workflow; joins, temporal semantics and faceting must be checked against explorEDA.

Capabilities to investigate. Joining by typed timestamps, separately scoped controls, shared scales and palettes, facet wrapping, reference layers that intentionally ignore facet filtering, and readable large-point rendering.

Graphical edits that should update text. Changing x/y fields, calendar category, facet layout, category order, palette, opacity, point size, bin width, labels or legend should change the defining text. Switching a chart's category should also update its grouping and labels rather than leaving stale calculations.

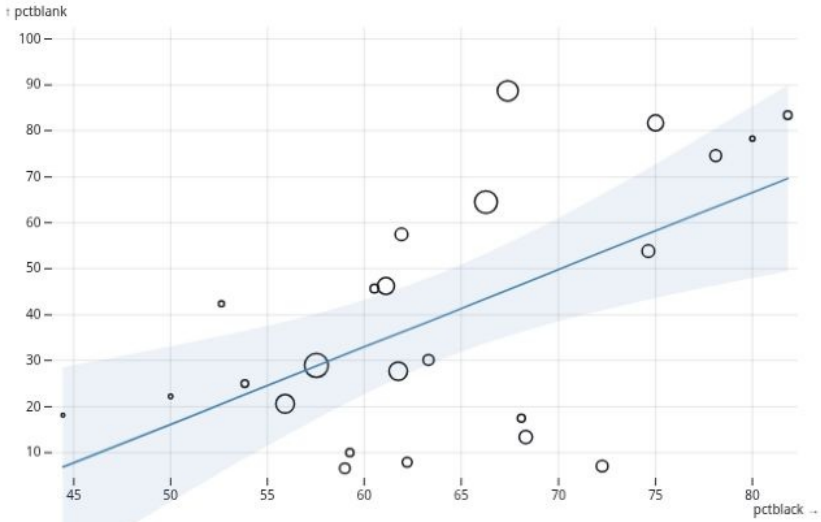
Control scope. Observed: the hour facet control changes the second result; the first remains month-colored. Source: palette is shared across both charts. Neither radio changes the upstream joined table. The archived source additionally lets the first period selector reverse the faceted Cividis colors when daytime is selected. An exact reproduction should preserve or explicitly revise this secondary styling dependency.

Execution and editing lesson. This tests a shared computed relation with separate interactive branches, plus a background result whose scope differs from the foreground. It also tests whether two-way editing preserves data meaning when calendar grouping changes.

1778	145	74.627	53.793
1792	128	72.269	7.031

We filtered out officers who stopped less than 10 people because their numbers could easily be swung by pure luck. Those officers accounted for less than 1% of the total stops recorded by the department.

When we graph the rest of the officers to show the percentage of stops where they failed to record race, versus the percentage of stops where they did record race and the driver was Black, we find that the officers who were more likely to leave tickets blank were also more likely to record that drivers they stopped were Black rather than White. This means that the percentage of stops involving Black drivers we see in cases where race was recorded is likely an undercount.



Where ticketed drivers live

Finally, we used the addresses in the ticket data to carry out a geospatial analysis of the places where drivers who received tickets from Bratenahl police stop come from.

Over 50% of drivers came from majority Black census tracts. About 11% of all

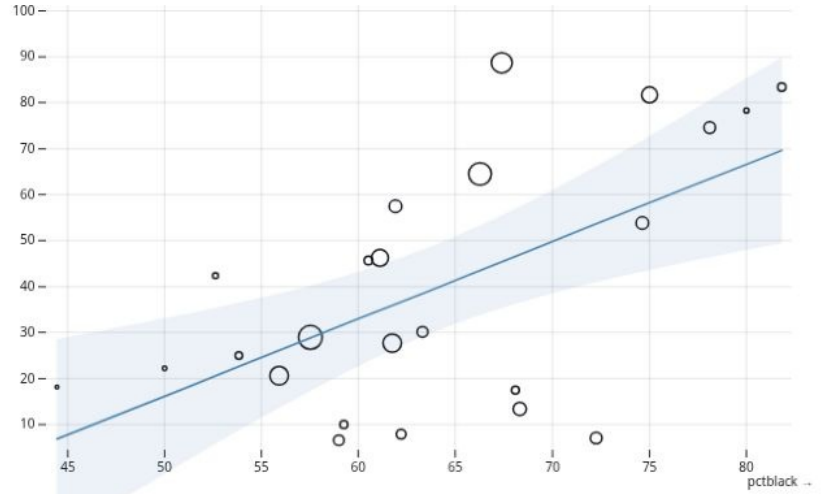
06. How Bratenahl, Ohio, Disproportionately Targets Black Drivers - Screenshot 1. The officer comparison places an interpreted grouped table, explanatory prose, a bubble scatter with regression and confidence band, and the next geographic question in one vertical reading sequence. Direct source. Browser side margins cropped; page content is unchanged.

Observable / Missing-data sensitivity > grouped rates > association > geographic corroboration

What the analysis does

The investigation asks whether Bratenahl's traffic enforcement disproportionately affects Black drivers and how incomplete race recording changes that assessment. It first examines missing data, then compares recording and race patterns across officers, and finally checks the neighborhoods where ticketed people live. In the rendered geographic tables, 498 of 956 tract-matched ticket records are associated with majority-Black tracts, and 776 with tracts below the stated Ohio median household income. These describe the matched records, not all stops or traffic exposure. The structure works because each section addresses a limitation or alternative explanation raised by the previous one, with prose interpreting the results and introducing a corroborating lens.

Role of prose. Prose establishes the reporting question, explains why two administrative sources answer different questions, makes missingness a methodological issue, interprets officer-level comparisons, and motivates geographic corroboration. It closes with reporting checks from court observations and sampled arrest records. This is one of the clearest examples of an argument requiring prose and results to remain in order.



Where ticketed drivers live

Finally, we used the addresses in the ticket data to carry out a geospatial analysis of the places where drivers who received tickets from Bratenahl police stop come from.

Over 50% of drivers came from majority Black census tracts. About 11% of all Ohio tracts are majority Black.

majority_black	stops	pct
true	498	0.521
false	458	0.479

Over 60% came from minority White census tracts. About 17% of all Ohio tracts are minority White.

minority_white	stops	pct
true	594	0.621
false	362	0.379

Over 80% came from tracts that are below the median Ohio household income. About 59% of all Ohio tracts are below the median household income. Bratenahl's median household income is about double that of Ohio.

06. How Bratenahl, Ohio, Disproportionately Targets Black Drivers - Screenshot 2. A later analytical step joins ticketed drivers' home locations to neighborhood characteristics, then presents successive grouped tables with interpretation. The notebook uses tables here rather than requiring a custom map. Direct source. Browser side margins cropped; page content is unchanged.

Source data and calculations

Original source. Bratenahl police stop records and ticket records obtained by the reporting team, supplemented with geocoded addresses and Census-derived tract attributes. The companion investigation was published by The Marshall Project and Cleveland reporting partners.

Row grain and scope. Stops: 4,048 police stop records, January 1, 2020-September 15, 2022. Tickets: 1,006 ticket records for January 1-September 15, 2022, with geocoded home locations. These are separate sources and grains. No deduplication proving one ticket row equals one unique driver was verified. Scope is Bratenahl enforcement, Ohio, with ticketed people's residences extending beyond the village.

Fields and units. Stops use date, loc1/loc2 (cross streets), dispo, sex, race and badge1. B and W are Black and White; blank and U are handled differently in different calculations. Other race codes are present but their meanings were not established in the inspected methods. Tickets contain initiating action and geocoding fields, including Longitude and Latitude in degrees. Public displays omit name/street address fields. Tract attributes include Black and White population percentages and median annual household income in USD.

- Stops CSV used by the notebook

- Geocoded ticket records used by the notebook

- Exact attached Ohio tract data

- Cleveland neighborhood geometry

- Municipality geometry

- Companion reported investigation

Calculations, dependencies and stated limits

- Overall missing race: (1,834 blank + 34 U) / 4,048 = 46.15%. The 2022 sensitivity branch filters dates after 2021-12-31, yielding 1,071 stops, and classifies both blank and U as missing: 354 missing, 717 recorded.

- Sensitivity control whitePct ranges 0-100 by five, initially 75. For missing-count N, the source adds round(whitePct/100 × N) to White and round((1 - whitePct/100 - 0.01) × N) to Black. Other categories are unchanged; shares divide by the sum of adjusted counts. Although the prose reserves one percent for other categories, the source does not explicitly add that remainder to them. Do not replace this with a denominator of all 1,071 without identifying it as a methodological change.

- Officer analysis counts by badge1 and race, pivots race codes into columns, and retains officers with total > 10. total includes blank + A + W + B + F + I + U. pctblack = 100 × B / (A + W + B + F + I + U); pctblank = 100 × blank / total. Thus U remains in the officer Black-share denominator, while the sensitivity branch treats U as missing. The prose's less-than-ten exclusion is slightly different from the actual greater-than-ten inclusion.

- The officer scatter uses x = pctblack, y = pctblank, radius = total and hover title = badge number. Plot.linearRegressionY fits the displayed regression with ci: 0.95. The code and rendered confidence band are visible; the fitted coefficient values are not reported.

- Ticket coordinates are parsed as numbers and converted to points. Each polygon receives the count returned by turf.pointsWithinPolygon. The tract collection filters attachment records to those with a type property, leaving 3,162 valid features from an array of 3,168.

- Tract flags are Black percentage >= 50; White percentage < 50; and median household income < 62,262 USD. Grouped tables sum ticket-point counts and divide each subtotal by the matched total. The inspected majority-Black table is 498/956; the income table is 776/956. Municipality and neighborhood matching have different denominators and should not silently reuse 1,006 or 956.

What feeds later results. Stops > date/missingness subsets > sensitivity estimate > associated percentages and prose. Separately, stops > badge/race counts > pivoted rates > officer table and scatter. Tickets > coordinate points + polygons > polygon counts > demographic flags/grouped shares > interpreted geography tables. The branches share some sources but are not one linear execution chain.

Reproduction sufficiency. The notebook successfully loads raw tables, the officer graph and geographic summaries; the source CSV URLs and tract attachment were inspected. These are sufficient for a focused officer comparison, and most of the larger investigation. Full geographic reproduction requires the supplied geometry, its recorded demographic attributes and the matching rules. The exact Census/ACS vintage and table identifiers were not established from the inspected body, so a current Census download is not an exact substitute. If the attachment disappears, use a documented historical tract release and report changed boundaries/vintages and resulting counts.

Limits and uncertainty. The sensitivity output currently renders SyntaxError: Unexpected token, although its calculation definitions can be inspected. Its dynamic output was therefore not verified by moving the slider. Other analytical branches work. Distinguish unknown-race policies, source populations and matched denominators. Tract racial composition is contextual evidence, not the individual driver's race; the analysis does not supply a traffic-exposure baseline establishing causation.

Elements needed to reproduce it

1. Establish the enforcement question > inspect stops and tickets separately > explain each source's omissions.

2. Quantify missing race > define a hypothetical allocation of missing cases > report sensitivity estimates and limitations. The current output is broken; source logic remains visible.

3. Ask whether officer recording practice explains the pattern > group by badge/race and compute rates > show table and bubble scatter > interpret association and sample-size limits.

4. Ask where ticketed people live > spatially match home coordinates > sum records by tract characteristics > compare majority-Black, minority-White and income categories.

5. Move from tracts to municipalities and Cleveland neighborhoods > inspect ranked tables > compare with reporting observations and acknowledge what the evidence establishes.

Element	Inputs	Calculation or configuration	Visible result	Interaction	Dependency
Essential: reporting question and limitations	Question; two administrative sources	Explain what each source records and why missingness matters	Narrative framing and transitions	None	Interpretations stay attached to their result and population
Essential: source previews	Stops CSV; tickets CSV for broader version	Parse dates/numbers; display selected fields and counts	Separate tables with grain and date scope	Table inspection observed	Distinct input branches; no implicit union
Essential: missing-race calculation	race and date	Explicit blank/U classification; 2022 subset; counts and shares	Counts, percentages and explanatory text	None needed for descriptive baseline	Feeds sensitivity branch; not automatically officer policy
Optional to minimum; essential to full sensitivity section	Observed race counts; missing count; whitePct	Apply inspected allocation formula; recompute adjusted totals/shares	Scenario table and dependent narrative; currently broken on original page	0-100 slider in steps of 5 observed; result update unverified	Only sensitivity estimate and its dependent prose
Essential: officer summary calculation and table	badge1, race	Group and pivot counts; calculate pctblack/pctblank; retain total > 10	Twenty-five officer groups and explicit denominators	Proposed minim um-count/filter edits	Shared derived table feeds graph
Essential: officer association chart	Officer rates and totals	x = Black share percent; y = blank share percent; radius = total; hover badge; linear fit with 95% interval	Full-width bubble scatter, axes and confidence band	Hover observed; threshold/control edits proposed	Same summary table as preceding output
Essential to full geographic argument	Ticket points; trac t/place/neighborh ood polygons and attributes	Count points per polygon; flag demographic/income categories; group sums and matched shares	Sequential category and ranked-location tables	No map or linked selection required	Point matching feeds every geographic comparison
Optional polish	Chart titles and selected interpretation	Percent axis labels, emphasis/callouts and consistent typography	Clearer reading and annotation	None required	Presentation; does not alter grouping

Fit for explorEDA

Smallest faithful version. Use the stops CSV, source/missingness explanation, officer grouped table and scatter with interpretation. This preserves a coherent section of the argument without requiring the whole spatial workflow. A larger faithful version adds the sensitivity branch and geographic tables, with the current source errors and formula discrepancy resolved explicitly.

Likely fit, subject to verification. The supplied product capabilities suggest a fit for table inspection, date filters, grouped calculations, calculated rates, an interactive scatter and saved settings. This remains a

hypothesis because explorEDA itself was not inspected.

Capabilities to investigate. Pivot or equivalent grouped aggregates, reuse of a derived table, regression and confidence intervals, size encodings, calculated values inside prose, branch-specific error handling, and point-in-polygon matching for the full geography section. None intrinsically requires arbitrary JavaScript execution.

Graphical edits that should update text. Graphical edits to axes, radius, hover fields, percentage labels, regression/interval settings and the minimum-stop threshold should update the defining text. A threshold edit affecting the source summary must update both table and graph. Race classification and denominator choices belong in explicit calculation text; a visual edit must not silently change them.

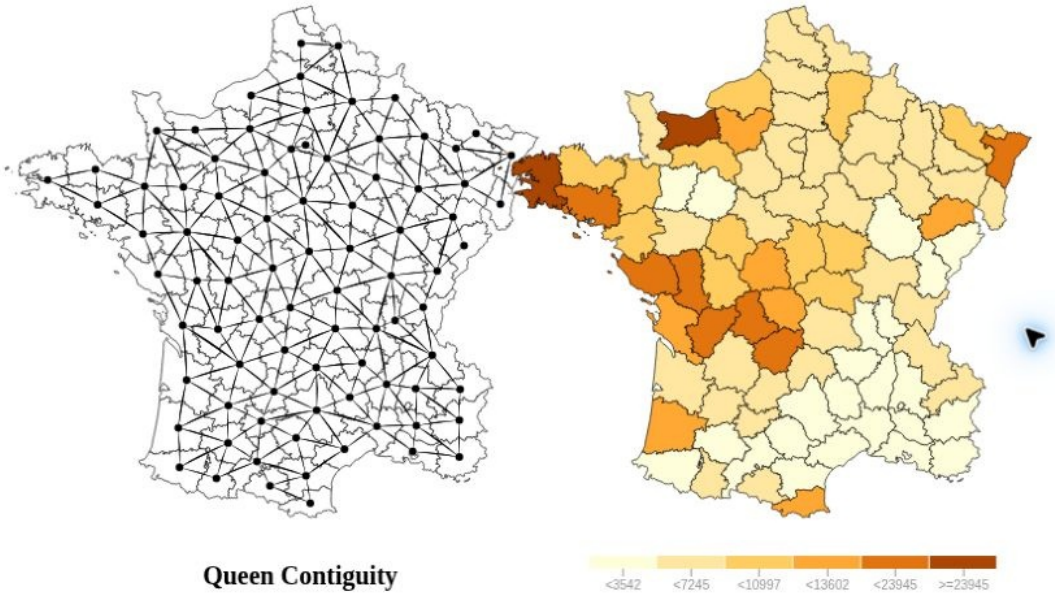
Control scope. The sensitivity slider is intended to affect its estimated table and accompanying numbers, not the independent officer or geographic comparisons. Officer threshold/filter changes should affect both its table and scatter. A geographic filter may affect several later tables only when they share the same matched dataset. No cross-chart linked selection was observed.

Execution and editing lesson. This tests a notebook's ability to preserve a public-facing analytical argument while calculations branch, denominators differ and one branch fails. Two-way editing must preserve numerical meaning across source text, table, graph and prose. The original's broken sensitivity output alongside working independent outputs is a concrete case for scoped invalidation and visible errors.

How do we determine who gets to be whose neighbor? **Contiguity.** Contiguity means two locations share a connection (using some criteria such as border or distance), that they follow a set of mathematical criteria that make them neighbors. Queen contiguity says that neighbors must share a common border or vertex. Rook says border only. **On a chess board, a square would be contiguous with squares of the opposite color under the rook criterion, while all 8 surrounding squares would count as neighbors under the queen criterion.** We'll be using queen contiguity weights as a demo, but feel free to experiment with other types.

Weight type Queen

```
conn = ▸ Object {arcs: Array(420), targets: Array(85), sources: Array(420)}
conn = geoda.getConnectivity(weights, data)                                    ojs
```



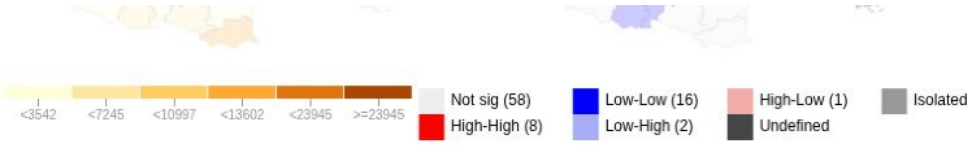
07. Local Spatial Autocorrelation Interactive Tutorial - Screenshot 1. The narrative explains contiguity, a Queen/Rook control changes the neighbor definition, and an explicit connectivity calculation precedes the maps. Direct source. Browser side margins cropped; page content is unchanged.

Observable / Neighbor definition > standardization > spatial lag > global/local inference

What the analysis does

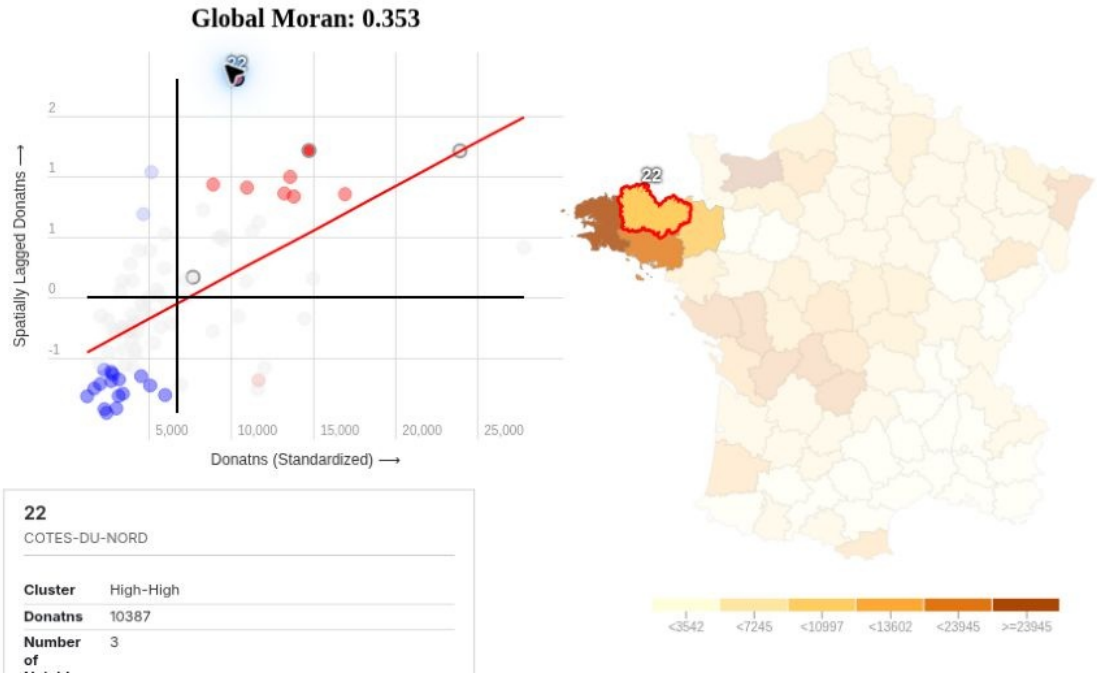
Using Guerry's charitable-donation data, the walkthrough asks which apparent concentrations in nineteenth-century France are statistically supported. It develops a global Moran statistic, then asks where the clusters occur. The illustrated low-low pattern lies in the south/southeast; a low-high example shows how one large neighbor can pull a neighborhood average upward. This is a worked statistical investigation, although its teaching purpose is prominent.

Role of prose. The prose defines each prerequisite, explains formulas and null-reference distributions, interprets quadrants and local classifications, then guides a linked selection back to its neighboring observations.



Putting it all together!

Below, the interactive map can demonstrate the conceptual link. Again, the colored points on the scatterplot represent significant locations. If you click on them, it will show you the core and the neighbors on the natural breaks map, clearly showing what High-High, Low-Low, Low-High, and High-Low clusters mean.



07. Local Spatial Autocorrelation Interactive Tutorial - Screenshot 2. Later linked investigation after selecting department 22, Côtes-du-Nord: the scatter, selected region and its neighbors connect the classification to its geographic context. Direct source. Browser side margins cropped; page content is unchanged.

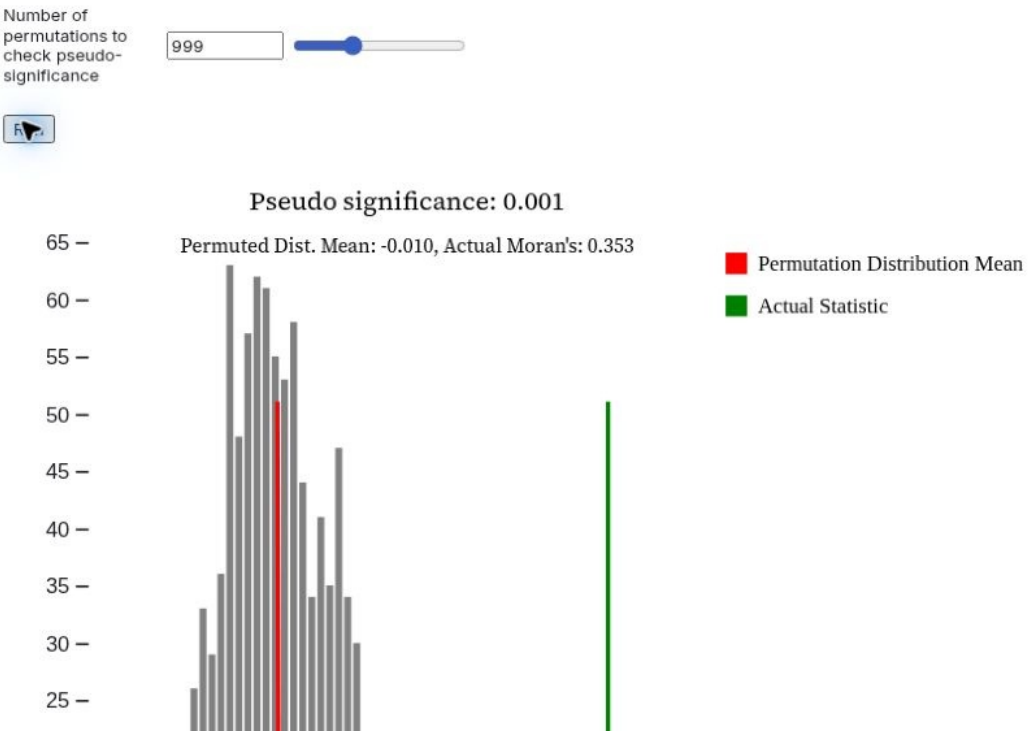
permutations, and for easy visualization a histogram of the permuted Moran statistics are generated. Permutation does not require the analytical p-value's assumptions about the data, which makes it useful for our interpretation.

The reference distribution created by the randomized values is used to calculate a so-called pseudo p-value:

$$p = \frac{R+1}{M+1}$$

where R is the number of times the computed Moran's I from the spatial random data sets (the permuted data sets) is equal to or more extreme than the observed statistic. M equals the number of permutations. The latter is typically taken as 99, 999, etc., to yield nicely rounded pseudo p-values.

Below, you will find a simulation of the calculation of the pseudo p-value.



07. Local Spatial Autocorrelation Interactive Tutorial - Screenshot 3. The explicit permutation rerun: explanation and the pseudo-p formula precede the Run control and reference-distribution result. This screenshot includes only the upper portion of the tall histogram. Direct source. Browser side margins cropped; page content is unchanged.

Source data and calculations

Original source. The GeoDa Center's Guerry collection, based on André-Michel Guerry's 1833 study and additional historical sources; an 85-department WGS84 GeoJSON is attached.

Row grain and scope. One of 85 historical French departments, with polygon geometry and 23 attributes. The collection spans 1815-1834; individual variables describe different periods rather than a common time series.

Fields and units. dept (stable identifier), Dprtmnt (name), geometry and Donatns (default variable). The tutorial calls Donatns charitable donations per capita; the original codebook specifies donations to the poor but does not resolve the monetary unit/denominator. Derived z scores and spatial lags are dimensionless.

- Exact attached Guerry GeoJSON

- Original dataset documentation

- Original downloadable Guerry ZIP

- Original researcher's codebook and CSV links

- Authored Local Moran methodology

- Authored global Moran methodology

Calculations, dependencies and stated limits

- Choose Queen (shared vertex/border) or Rook (shared border) adjacency; row-standardize neighbor weights for weighted averages.

- Visible standardize function uses ss.mean, ss.standardDeviation and ss.zScore; calculate spatial_lags with geoda.spatialLag(weights,std_vals).

- Fit the spatial lag on standardized values; the source takes bestFit.slope as global Moran I. The observed default is approximately 0.353.

- Generate permutation reference distributions and pseudo-p=(R+1)/(M+1). Local conditional permutations fix the selected location and shuffle other values. The live Local Moran call supplies weights, selected column, permutation count, lookup method and significance cutoff.

- Local Moran values, pseudo-p values and quadrants feed class colors, counts, maps and selection details. Six natural-break classes provide the original-value map.

What feeds later results. Geometry > weights/connectivity; chosen variable > z scores > neighbor lags > scatter/global I. Weights + variable + permutations + threshold > local statistics > class counts/maps. Department selection links detail statistics, scatter and geographic neighbors.

Reproduction sufficiency. Core methods and data are exposed, with substantive primary methodological documentation. No random-seed control was observed, so exact permutation outcomes should not be promised. Clicking Run changed the null-distribution mean from -0.013 to -0.010 while I remained 0.353 and pseudo-p 0.001.

Limits and uncertainty. Two live standalone cells display SyntaxError (the initial distribution/histogram cell and a standalone LISA-map cell), while the main neighbor map, permutation histograms, Moran scatter and later linked views render. The final linked scatter labels its x-axis standardized but displays raw-scale ticks; use the earlier correct standardized scatter for the analytical reproduction. The source workbook discusses multiple-comparison limitations; local classes are exploratory and depend on threshold/permutations.

Elements needed to reproduce it

1. Define the hotspot question > load departmental data > choose and inspect neighbor weights.

2. Choose the donation field > standardize > compute neighbor lag > explain the Moran scatter and global statistic.

3. Construct a permutation reference > compare observed I with randomized results > ask where clustering occurs.

4. Compute local statistics and classes > inspect a cluster map/table > select a department and examine its neighbors > interpret the relationship.

Element	Inputs	Calculation or configuration	Visible result	Interaction	Dependency
Essential: explanatory text and formulas	Question; definitions; selected field	Explain weights, z scores, lag, global/local statistics and pseudo-p	Formula and interpretation blocks	None	Each prerequisite supports the next calculation
Essential: source and neighbors	Department geometry/id	Queen or Rook adjacency; normalized neighbor weights	Inspectable neighbor relation; optional contiguity map	Weight-type selector	Feeds all spatial calculations
Essential: standardized comparison	Donatns; mean/sd; neighbors	z=(x-mean)/sd; lag=weighted neighbor average; x=z; y=lag; zero rules; fitted slope	Moran scatter and I summary	Variable selector	Upstream field/weights affect multiple later blocks

Element	Inputs	Calculation or configuration	Visible result	Interaction	Dependency
Essential: inference result	Observed statistic; permutation count	Generate reference statistics; histogram and observed-value rule; pseudo-p formula	Null distribution and significance summary	Permutation count; explicit Run	Randomized result should retain execution state
Essential: local results	Weights, field, permutations, cutoff	Local Moran; conditional pseudo-p; quadrant plus significance classification	Department result table and class map/legend	Significance and permutation controls	Feeds counts, scatter colors and maps
Optional: linked geographic explanation	Selected department and neighbors; local results	Highlight core and neighbors; show field, class, local statistic and p-value	Linked map/scatter and detail panel	Click/hover; department-ID selector	Observed selection of dept22 exposed its High-High classification and 3 neighbors
Optional: natural-break comparison	Original donation values	Six-class map beside local classification map	Original intensity versus spatial class	Linked selection	Can be presented as separate full-column blocks

Fit for explorEDA

Smallest faithful version. Fix the Guerry field and Queen neighbors, retain the calculation chain and formulas, show the correct standardized scatter/I, a permutation histogram, and a local-results table plus one map. The bespoke linked two-map apparatus can be omitted. Precomputed statistics would reproduce a static story but would not test recalculation.

Likely fit, subject to verification. Text, tables, calculated columns, controls, saved state and ordinary charts plausibly cover the presentation and inspection pieces. Spatial/statistical computation is a capability to investigate, not assumed.

Capabilities to investigate. Neighbor relations or geometry support, grouped weighted calculations, global/local spatial statistics, randomized execution with repeatable state, explicit rerun controls, linked selection across blocks, and error propagation without obscuring functioning results.

Graphical edits that should update text. Axes and field mappings, thresholds, permutation count, binning, reference lines, class colors, map classification and selection keys should persist to text. A graphical change to a significance cutoff changes the derived classification, not merely its styling.

Control scope. Weights and variable affect many downstream blocks; local significance/permutation settings affect local results and colors. A selected department changes the linked explanation and local simulator. The global Run refreshes its randomized reference result.

Execution and editing lesson. This tests a deep dependency chain, the difference between changing a parameter and rerunning a stochastic calculation, and saving enough state to review a result. It also exposes the need to keep formulas, units, labels and dependent graphics synchronized.

In [2]:
```
# Load the data file, and create a column with total north/south traff

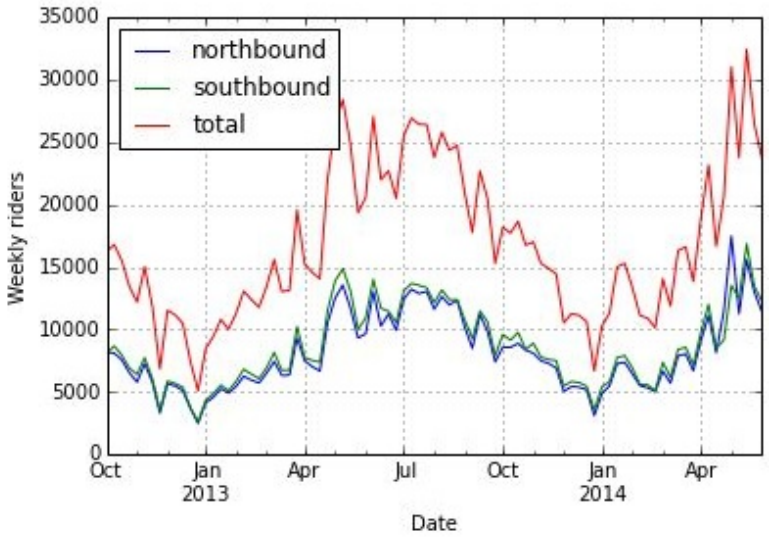
hourly = pd.read_csv("FremontHourly.csv", index_col='Date', parse_date
hourly.columns = ['northbound', 'southbound']
hourly['total'] = hourly['northbound'] + hourly['southbound']
```

In [3]:
```
# Resample the data into daily and weekly totals

daily = hourly.resample('d', 'sum')
weekly = daily.resample('w', 'sum')
```

Now let's take a peek at our data and see what it looks like:

In [4]:
```
weekly[['northbound', 'southbound', 'total']].plot()
plt.ylabel('Weekly riders');
```



The red line shows the total number of weekly crossings, which is the sum of the northbound and southbound crossings.

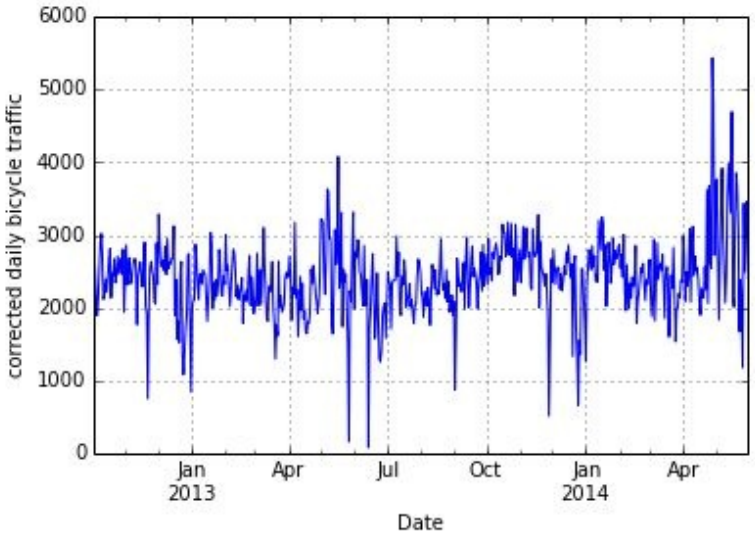
At first glance, April and May 2014 include some spikes in the data: over 32,000 riders per week crossed the bridge one week in May! This trend might be a bit clearer if we use a **moving window average**: basically, for each day we'll take the average of the 30-day period around it:

08. Is Seattle Really Seeing an Uptick In Cycling? - Screenshot 1. The opening data examination: Python aggregation cells, weekly directional totals and prose introducing a moving-average comparison in one reading column. Direct source. Geometric excerpt of the captured page; the uncropped capture is included in the ZIP.

This shows the data along with the overall trend, accounting for precipitation and temperature extremes. Let's de-trend the data with this model and see what's left over:

In [22]:
```
daily['overall_corrected'] = daily['total'] - daily['overall_trend'] +
print("rms = {0:.0f}".format(np.std(daily['overall_corrected'])))
daily['overall_corrected'].plot()
plt.ylabel('corrected daily bicycle traffic');
```

```
rms = 520
```



From the reduced RMS, we can see that our model does a fairly good job capturing the trends. But there's still some daily variation that it cannot account for. We could think about what this might be, and try to capture it with more inputs to the model. For example, a person's choice to ride might be affected not just by the weather today, but by the weather yesterday. We also could account for freezing temperatures, which may manifest as a nonlinear effect.

In order to better see the remaining trend, let's plot a month-long moving-window average over the corrected data:

08. Is Seattle Really Seeing an Uptick In Cycling? - Screenshot 2. A later dependent step: the correction formula, residual-variability output, corrected traffic chart and interpretation introducing the next rolling calculation. Direct source. Geometric excerpt of the captured page; the uncropped capture is included in the ZIP.

Python notebook / author blog / Time series > confounder adjustment > residual trend

What the analysis does

The investigation asks whether apparent growth in Fremont Bridge bicycle traffic survives adjustment for daylight, weekday, and weather. It develops an author-reported historical estimate of roughly 10% annual growth in **bridge crossings**, not distinct cyclists or citywide ridership. Each paragraph explains why the preceding view is insufficient and introduces the next covariate or calculation. Visible Python cells make the progression from raw counts to adjustments unusually clear. The two captured regions demonstrate full reading-column blocks rather than a dashboard.

Role of prose. Each paragraph explains a limitation of the preceding view, introduces the next confounder or calculation, and interprets the resulting correction. Code cells remain visible between these decisions and

their plots.

Source data and calculations

Original source. Original data: Seattle's Fremont Bridge counter dataset and NOAA daily observations for SeaTac Airport station USW00024233. The article links NOAA's daily data search; that old service endpoint was not independently verified. The author's exact CSV snapshots were fetched and inspected.

Row grain and scope. Crossing grain: one hour at one bridge, Date plus northbound and southbound count. Snapshot: 14,568 rows, October 2, 2012 through May 31, 2014. Counts are passages. Some directional counter cells are blank; the article has no explicit imputation block. Weather grain: station-day, 609 rows October 1, 2012 through June 1, 2014. Relevant fields DATE, TMIN and TMAX (tenths of °C), PRCP (tenths of mm). Other weather fields are unused. Code converts temperature to °F with 0.18*T+32 and rain to inches with PRCP/254, then joins by date.

Fields and units. Crossing data: Date and Fremont Bridge NB/SB, hourly crossing counts; calculated total, day/week and weekday. Weather: DATE, TMIN/TMAX in tenths of °C, PRCP in tenths of mm, converted to °F/inches. Calculated daylight hours and elapsed-day count enter successive models.

- Hourly crossing CSV

- weather CSV

- source repository

- downloadable notebook

- Fremont Bridge counter dataset

- NOAA's daily data search

Calculations, dependencies and stated limits

- Verified calculations: total = northbound + southbound; daily and weekly sums; 30-observation rolling mean; mean traffic by weekday; daylight-hours calculation from date, latitude 47.61° and axial tilt 23.44°; successive ordinary least squares models; predicted traffic; corrected series actual - prediction + mean(prediction). Later models add weekday indicators, daylight, minimum/maximum temperature, precipitation, then elapsed day count. The time coefficient is annualized relative to mean daily traffic.

What feeds later results. Dependency chain: hourly > daily > weekly/weekday summaries; date > daylight; daily + weather + weekday/daylight > fitted model > predictions > corrected traffic > 30-day rolling summary and final trend.

Reproduction sufficiency. The data and visible methods are sufficient to reconstruct the exploratory sequence, with explicit decisions about missing counter values, window alignment, legacy pandas behavior, and statistical uncertainty.

Limits and uncertainty. Method caveats: prose describes a centered 30-day mean, but the code does not request centering; reproduce the code's right-aligned/trailing setting explicitly, or label a deliberate centered variant. Old pandas documentation indexing confirms center=False default; the exact archived HTML/PDF page could not be reopened, so avoid claiming an archived-document inspection. Current official rolling documentation likewise defines right-aligned default. The source's explicit coefficient covariance inversion includes all seven weekday dummy columns plus an intercept, making the design singular/unstable; a temperature-error calculation also omits covariance. Do not present the printed ±2% as independently validated. Fit/prediction views remain a strong workflow reference; uncertainty estimation requires review.

Elements needed to reproduce it

1. question

2. load/inspect counts

3. total and temporal aggregation

4. weekly lines

5. seasonal interpretation

6. daylight/weekday calculations

7. comparison and fitted line

8. weather join and unit conversion

9. adjusted predictions

10. residual/rolling result

11. qualified growth interpretation

Element	Inputs	Calculation or configuration	Visible result	Interaction	Dependency
Question and caveats	Research question, source scope	Explain counts versus people and possible confounders	Reading-column prose	None observed	None
Count preparation	Hourly directional CSV	Parse date; NB+SB; sum per day/week	Small table and reusable daily/weekly results	Proposed date filter	Source
Weekly lines	Weekly NB, SB, total	x=week, y=crossings, color=direction/total; legend	Three-line time series	Proposed series/axis edits	Aggregated counts
Rolling summary	Daily total	Mean over 30 consecutive daily observations; alignment explicit	Smoothed time series	Proposed window control	Daily total
Weekday/daylight summaries	Dates and daily/weekly counts	Weekday means; calculated daylight; regression	Weekday comparison, daylight scatter and fitted line	Proposed groupi ng/predictor controls	Dates, daily/weekly
Weather preparation	Weather CSV, daily totals	Unit conversion and date join	Weather lines and model input table	Proposed date scope	Sources
Fitted and corrected series	Joined inputs, predictors	Fit; predict; actual-predicted+m ean(predicted)	Actual/expected and corrected time series	Proposed model specification control	Model inputs
Final interpretation	Corrected series, time coefficient	30-day summary; annualized trend; qualified uncertainty	Curve plus explanatory conclusion	No original widgets observed	Fit and corrected data
Essential versus optional	Reference scope	interleaved explanations, one transparent temporal aggregate, a visible correction model, predictions/corrected values, clear dependency names. Optional polish/full article: every successive model, solar formula as an editable block (a documented daylight input can substitute), exact Matplotlib styling, confidence intervals.	Faithful reduced version versus polish	As described above	Do not require omitted custom features for the core

Fit for explorEDA

Smallest faithful version. Weekly directional and total trends > a daily smoother and weekday summary > one weather-adjusted model and corrected trend, with explicit source scope and qualified interpretation. The model is needed to preserve the adjustment argument; stopping at the first time series would lose it.

Likely fit, subject to verification. Existing charts, tables, filters, calculated fields and saved state appear useful, based only on the product description.

Capabilities to investigate. Date joins, ordered-window calculations, named intermediate datasets, fitted-model outputs and coefficient/result tables. Daylight can be supplied as a documented derived input for a smaller version. Statistical uncertainty requires methodological review because of the source covariance problems.

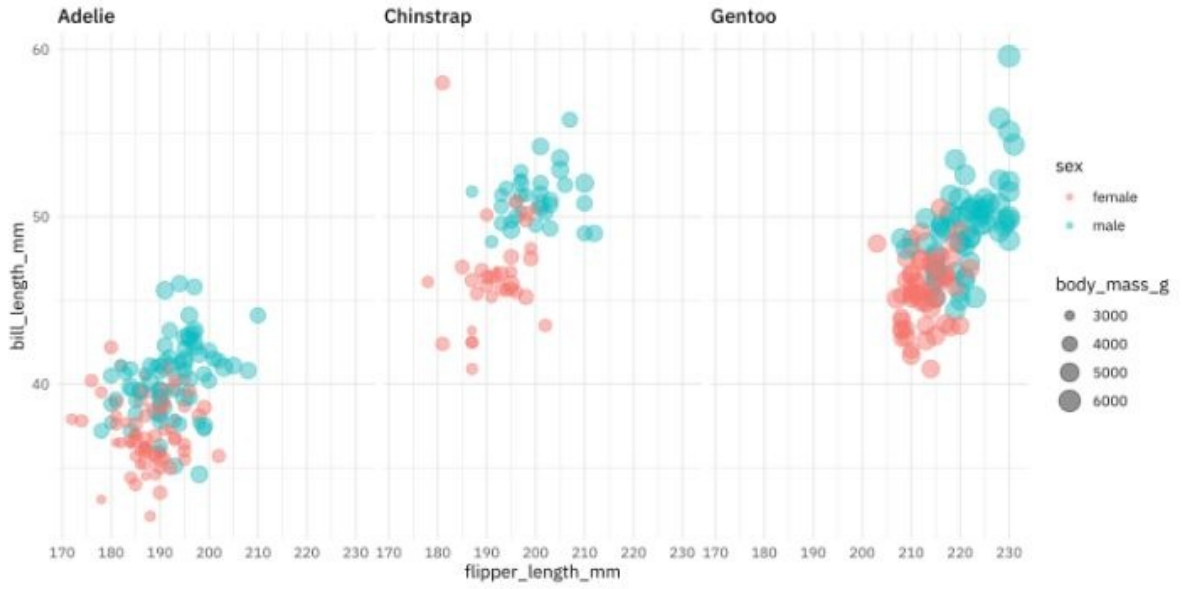
Graphical edits that should update text. Selected series, x/y encodings, temporal aggregation, window length and alignment, colors, legends, labels and axis limits should update the corresponding defining text. A chart appearance edit should not refit its model.

Control scope. A source date filter affects every dependent summary/model/output. A smoother window affects only its smoother and explicitly dependent statements. Any proposed predictor control must name the model it changes; the original article has static outputs rather than analytical widgets.

Execution and editing lesson. Named intermediate results and selective re-execution must preserve a long dependency chain while keeping prose attached to the result it interprets. No arbitrary-JavaScript requirement follows from this example.

If you try building a classification model for `species`, you will likely find an almost perfect fit, because these kinds of observations are actually what distinguish different species. Sex, on the other hand, is a little messier.

```
penguins %>%
  filter(!is.na(sex)) %>%
  ggplot(aes(flipper_length_mm, bill_length_mm, color = sex, size = body_mass_g)) +
  geom_point(alpha = 0.5) +
  facet_wrap(~species)
```



It looks like female penguins are smaller with different bills, but let's get ready for modeling to find out more! We will not use the island or year information in our model.

```
penguins_df <- penguins %>%
  filter(!is.na(sex)) %>%
  select(-year, -island)
```

Build a model

We can start by loading the tidymodels metapackage, and splitting our data into training and testing sets.

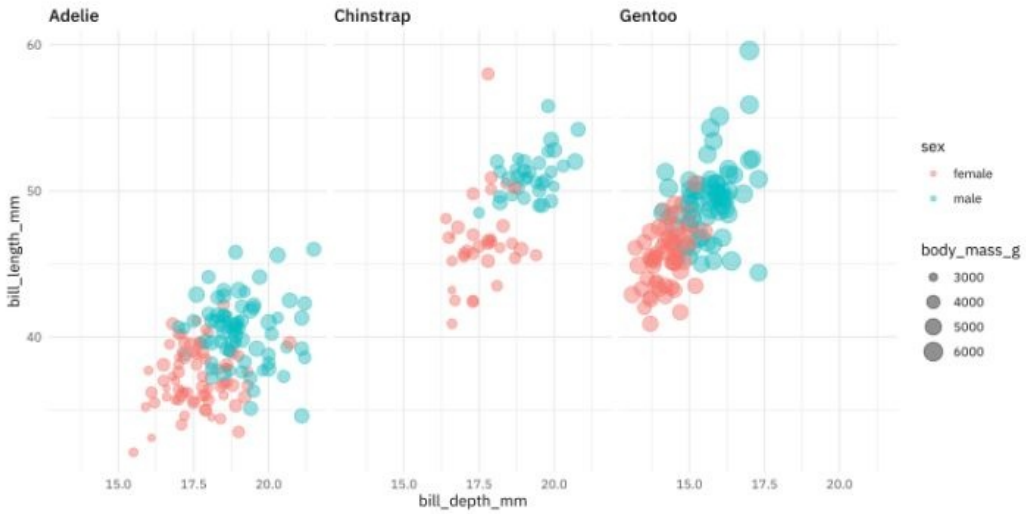
09. Get started with tidymodels and #TidyTuesday Palmer penguins - Screenshot 1. The first exploration: known-sex filtering and scatterplot code, a species-faceted result and interpretation before modeling. Direct source

```
penguin_final$.workflow[[1]] %>%
  tidy(exponentiate = TRUE)
```

```
## # A tibble: 7 x 5
##   term              estimate std.error statistic       p.value
##   <chr>                <dbl>     <dbl>     <dbl>         <dbl>
## 1 (Intercept)       3.12e-35  13.5         -5.90 0.00000000369
## 2 speciesChinstrap  1.34e- 3   1.70        -3.89 0.000101
## 3 speciesGentoo     1.08e- 4   2.89        -3.16 0.00159
## 4 bill_length_mm    1.78e+ 0   0.137        4.20 0.0000268
## 5 bill_depth_mm     3.89e+ 0   0.373        3.64 0.000273
## 6 flipper_length_mm 1.07e+ 0   0.0538       1.31 0.189
## 7 body_mass_g       1.01e+ 0   0.00108      4.70 0.00000260
```

- The largest odds ratio is for bill depth, with the second largest for bill length. An increase of 1 mm in bill depth corresponds to almost 4x higher odds of being male. The characteristics of a penguin's bill must be associated with their sex.
- We don't have strong evidence that flipper length is different between male and female penguins, controlling for the other measures; maybe we should explore that by changing that first plot!

```
penguins %>%
  filter(!is.na(sex)) %>%
  ggplot(aes(bill_depth_mm, bill_length_mm, color = sex, size = body_mass_g)) +
  geom_point(alpha = 0.5) +
  facet_wrap(~species)
```



Yes, the male and female penguins are much more separated now.

▶ **Details**

09. Get started with tidymodels and #TidyTuesday Palmer penguins - Screenshot 2. The later model result: coefficient table and interpretation motivate changing the scatterplot x-axis from flipper length to bill depth while retaining the other encodings. Direct source

R analytical walkthrough / Explore > fit and validate > revise the visual encoding

What the analysis does

Can body measurements and species predict a penguin's recorded sex? The article begins with an exploratory plot, compares two classification models, checks the selected model on held-out data, then uses its coefficients to revisit the visualization. Its argument has a useful return to an earlier analytical choice: bill depth separates recorded sexes better than the original x variable, after the model highlights that measurement. The prose gives each modeling step a purpose and distinguishes apparent separation from validation. It is a full code-and-result walkthrough, despite the introductory title.

Role of prose. Prose gives each modeling step a purpose, interprets held-out results and coefficients, then explains why an earlier axis choice should change. This distinguishes a visually plausible separation from a validated prediction and makes a concrete editing decision part of the argument.

Source data and calculations

Original source. Original collectors: Kristen Gorman and Palmer Station LTER. Original species files: Adelie, Gentoo, Chinstrap. These links were obtained from the authors' data documentation; the tidy CSV itself was fetched and parsed.

Row grain and scope. Row = one measured penguin, 344 records, three species, Biscoe/Dream/Torgersen islands in Antarctica's Palmer Archipelago, 2007-2009. Fields species, island, bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g, sex, year. Lengths in mm; body mass in grams.

Fields and units. species, island, bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g, sex, year. Four numeric body measurements enter the model with species; island and year are dropped.

- Tidy CSV

- repository file

- data documentation

- original data download documentation

- Adelie

- Gentoo

- Chinstrap

Calculations, dependencies and stated limits

- Filter missing sex: 333 usable records (168 male, 165 female), all complete on the four numeric features. The model drops island and year. The initial plot maps x=flipper length, y=bill length, color=sex, size=body mass, alpha=0.5, facet=species.

- Seed 123; stratified train/test split by sex, visible output 250 training and 83 testing records. Twenty-five bootstrap resamples of training data compare a logistic GLM and ranger random forest, predicting sex from species plus four measurements. Resampling results are similar; the author selects the simpler logistic model.

- Held-out model result reported by article: accuracy 0.940, ROC AUC 0.991. These are reproduced article outputs, not our newly validated estimates. Exponentiating coefficients gives odds ratios; the article interprets bill depth as particularly informative while flipper length has little additional evidence conditional on the other predictors.

- The concluding plot changes x to bill depth and keeps the other encodings. Source code for all calculations and data is visible/available. Exact old-package random samples/results can vary with software versions.

What feeds later results. Source > known-sex rows > first and revised plots. Known-sex rows > seeded stratified split > training resamples > model comparison/selection > held-out predictions/metrics and coefficients. The coefficient interpretation motivates a visual revision, rather than mechanically changing the data.

Reproduction sufficiency. The tidy CSV was fetched and parsed, with all 333 known-sex rows complete on the four numeric measurements. Inline R exposes all preparation, modeling and plotting steps; original species data links are documented.

Limits and uncertainty. The target is recorded sex, not species. Article model scores are reported outputs, not new validation by this research. Exact old-package resamples/results may differ with software versions. The optional video is not needed to recover the visible analysis.

Elements needed to reproduce it

1. question/data context
2. known-sex filter
3. first scatter and interpretation
4. split/resampling definitions
5. two model results
6. choose and test model
7. coefficient table and interpretation
8. revised scatter

Element	Inputs	Calculation or configuration	Visible result	Interaction	Dependency
Context and data preparation	Penguin records	Explain measured variables; exclude unknown sex	Prose, counts, data table	Proposed species filter	Source
Exploratory scatter	Known-sex rows	x=flipper length; y=bill length; sex color; mass size; species facet	Three-facet scatter	No original analytical widget; proposed encoding edits	Filtered rows
Essential: training/held-out partition	Filtered rows, seed	Drop year/island; stratified 250/83 split	Partition summary	Proposed seed/split control	Data scope
Optional: model comparison	Training partition	25 bootstraps; logistic and forest metrics	Accuracy/AUC table; confusion summary	Proposed model selection	Training data
Essential: held-out validation	Selected fitted model, held-out rows	Apply the trained model to held-out rows; compute accuracy and confusion counts	Small performanc e/confusion table; ROC curve is optional	None required; threshold control is optional and not observed	Fit and held-out partition
Coefficients	Logistic fit	Exponentiate estimates; interpret effects	Odds-ratio table plus prose	Optional display ed-coefficient selection	Fit
Revised scatter	Same filtered records	x=bill depth; other encodings retained	Stronger visible separation	Proposed x-axis edit	Filtered data, editorial motivation from coefficients
Essential versus optional	Reference scope	Essential: first plot, transparent filtered sample, one training/held-out partition, logistic fit, held-out accuracy/confusion table, interpreted coefficient output and revised plot. Optional: random-forest comparison, 25 bootstraps, ROC traces, video and exact theme. Facets can be three vertically consecutive species charts.	Faithful reduced version versus polish	As described above	Do not require omitted custom features for the core

Fit for explorEDA

Smallest faithful version. Known-sex filter > first scatterplot > one stratified training/held-out split > one logistic fit > a small held-out accuracy/confusion table > coefficient table with interpretation > revised scatterplot. This preserves a genuine prediction check; the random-forest comparison and 25-bootstrap exercise are optional.

Likely fit, subject to verification. Filtering, interactive scatterplots, size/color controls, tables and calculated fields appear compatible with the stated capabilities. Facets could be three vertically consecutive species charts if needed.

Capabilities to investigate. Faceting, logistic model fitting, predicted-class/probability outputs, held-out partitions, seeded resampling and model-result tables. The app description does not establish these capabilities.

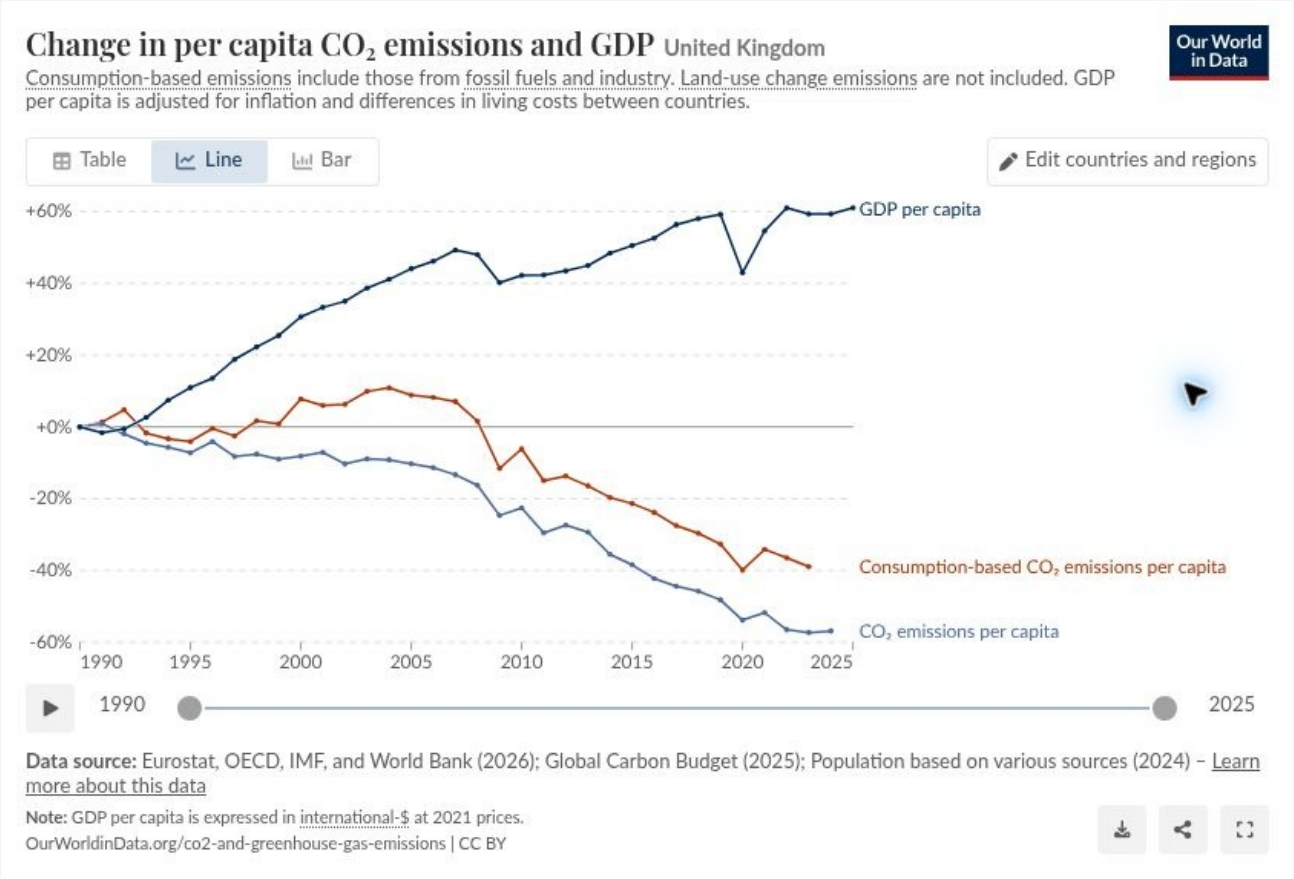
Graphical edits that should update text. Graphical edits to x/y fields, color, size, opacity, facets and labels should update the block text. Changing a chart x-axis should not silently redefine the model predictor set.

Control scope. Data-scope changes affect both visualizations and all model/validation descendants. A visual encoding change affects only that output. The original analytical walkthrough has no live model or chart controls; these editing controls are proposed.

Execution and editing lesson. A model result can motivate revising an earlier chart without creating an execution cycle: both charts still depend on the filtered records, while the model provides editorial motivation. Keep analytical model definitions distinct from chart encodings.

decoupled the two.

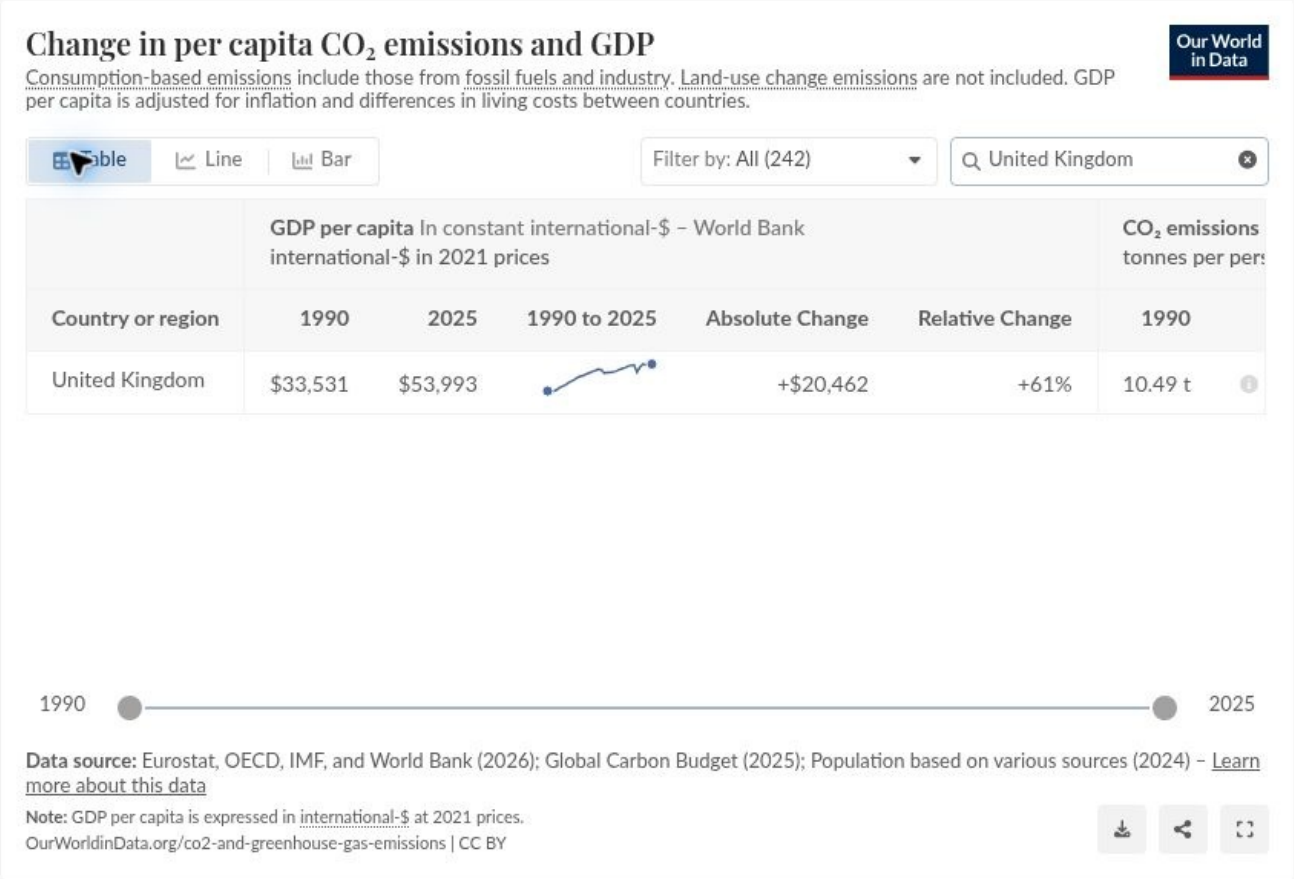
Take the UK as an example. It is shown in the chart. This chart shows the change in GDP and annual CO_2 emissions per capita since 1990. We see that the UK's GDP has increased a lot over the last 30 years while its emissions have fallen. You can also see the data without per capita adjustments.



It's not just the UK. Many other countries have achieved this decoupling. Using the "Edit countries and regions" button on the chart, you can see this for yourself. France, Germany, Sweden, Finland, Denmark, Italy, Czechia, and Romania are examples of countries where we see this.

10. Many countries have decoupled economic growth from CO_2 emissions, even if we take offshored production into account - Screenshot 1. The UK comparison: explanatory text, a three-series percentage-change chart with source notes and controls, followed by the next-country question. Direct source

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10. Many countries have decoupled economic growth from CO_2 emissions, even if we take offshored production into account - Screenshot 2. An interaction actually performed: switch the same block to Table and search for United Kingdom. The table exposes raw units and percentage changes while retaining the narrative context. Direct source

Data journalism / OWID / Normalize time series > compare definitions > compare countries

What the analysis does

The article tests whether GDP can increase while carbon emissions fall, then considers the objection that a country may simply have moved emissions abroad. The UK comparison puts economic output, territorial emissions and consumption-based emissions on a common relative-change scale. The narrative then asks readers to inspect other countries before distinguishing national progress from inadequate global progress. Prose defines the metrics, anticipates an objection, interprets the result, and guides the next comparison. The main two figures are full-width sequential blocks; the later country-facet image is optional, not a dashboard requirement.

Role of prose. Prose defines production and consumption emissions, anticipates the objection that emissions have moved abroad, interprets the UK result, asks readers to inspect other countries, and separates national progress from the global conclusion.

Source data and calculations

Original source. Original sources: World Bank GDP per capita PPP indicator, sourced to Eurostat/OECD/IMF/World Bank (2026); Global Carbon Budget (2025); OWID population sources (2024). Download metadata explicitly documents per-capita calculation and carbon-to-CO_2 conversion.

Row grain and scope. Row = entity-year; current download has 8,622 rows, 242 countries/regions, years 1990-2025. Fields Entity, Code, Year, GDP per capita, CO_2 emissions per capita, Consumption-based CO_2 emissions per capita. Nulls and series endpoints differ.

Fields and units. GDP unit is international dollars at 2021 prices (inflation and purchasing-power adjusted). Emissions are tonnes CO_2/person, fossil fuels and industry; land-use change is excluded. Consumption emissions = production emissions - embedded exports + embedded imports, then divide by population.

- chart/data page

- Full chart CSV

- metadata JSON

- World Bank GDP per capita PPP indicator

- Global Carbon Budget

- OWID population sources

Calculations, dependencies and stated limits

- GDP unit is international dollars at 2021 prices (inflation and purchasing-power adjusted). Emissions are tonnes CO_2/person, fossil fuels and industry; land-use change is excluded. Consumption emissions = production emissions - embedded exports + embedded imports, then divide by population.

- Displayed normalization: relative change from 1990. Our explicit check 100*(value_t/value_1990-1) on downloaded values agrees with the displayed UK endpoints: GDP +61.02% (2025), territorial emissions -56.87% (2024), consumption emissions -38.91% (2023). Those endpoints are different years; do not describe all three as a 2025 comparison. Exact Grapher behavior for missing baseline data was not inspected.

What feeds later results. Entity-year prepared metrics > country selection > per-series 1990 baseline lookup > normalized series > line comparison/table. Repeat for other countries using the same source/definitions. Preserve the raw levels separately from percentage-change outputs.

Reproduction sufficiency. The actual full-data download, metadata and README were inspected, and the relative-change calculation was checked against the displayed UK endpoints. The exact prepared dataset and documented definitions suffice for the article-level investigation. Reconstructing the underlying trade-embedded emissions estimates from raw trade data is outside this reference.

Limits and uncertainty. Displayed normalization: relative change from 1990. Our explicit check 100*(value_t/value_1990-1) on downloaded values agrees with the displayed UK endpoints: GDP +61.02% (2025), territorial emissions -56.87% (2024), consumption emissions -38.91% (2023). Those endpoints are different years; do not describe all three as a 2025 comparison. Exact Grapher behavior for missing baseline data was not inspected. The newer article charts should be reproduced from this exact download, not automatically from the convenient owid/co2-data repository: its GDP series/units can differ.

Elements needed to reproduce it

1. decoupling question

2. definitions and UK selection

3. baseline calculation

4. three-series line chart

5. interpretation and offshoring objection

6. consumption-based comparison

7. other-country selection/comparison

8. limits/global context

Element	Inputs	Calculation or configuration	Visible result	Interaction	Dependency
Metric definitions	Source metadata	Explain production/consumption, per capita, PPP and exclusions	Prose and source notes	Links observed	None
Prepared annual data	Chart CSV	Preserve nulls and entity-year keys	Inspectable raw table	Table tab/search observed	Source
Baseline transform	Selected entity, 1990 values	Percentage change from baseline; source values retained	Three normalized series	Time range control observed; baseline edit proposed	Entity data
UK line chart	Normalized GDP and both CO_2 series	x=year; y=% change; color=metric; zero rule; line-end labels	Decoupling evidence	Table/Line/Bar, country edit, timeline, downloads observed	Baseline transform
Interpretation	Result and definitions	Explain divergence and trade-adjusted comparison	Prose	None	Analytical result
Broader comparison	Other selected countries	Same definitions and normalization	Next country comparison	Country editing observed	Shared source and transform
Caveats	Coverage and national/global distinction	Show actual series endpoints and excluded emissions	Prose/notes	Optional source details	Metadata/results
Essential versus optional	Reference scope	three measures, explicit baseline and units, country selection, separate interpretation blocks, source/coverage note. Optional polish: playback, country facet grid, download dialog styling, sparkline table cells.	Faithful reduced version versus polish	As described above	Do not require omitted custom features for the core

Fit for explorEDA

Smallest faithful version. UK raw-data table > three baseline-normalized lines > interpretation > repeat for a second country. Preserve both territorial and consumption emissions to keep the offshoring question intact.

Likely fit, subject to verification. Calculated fields, charts, tables, filters and saved analysis state appear sufficient for much of the prepared-data workflow.

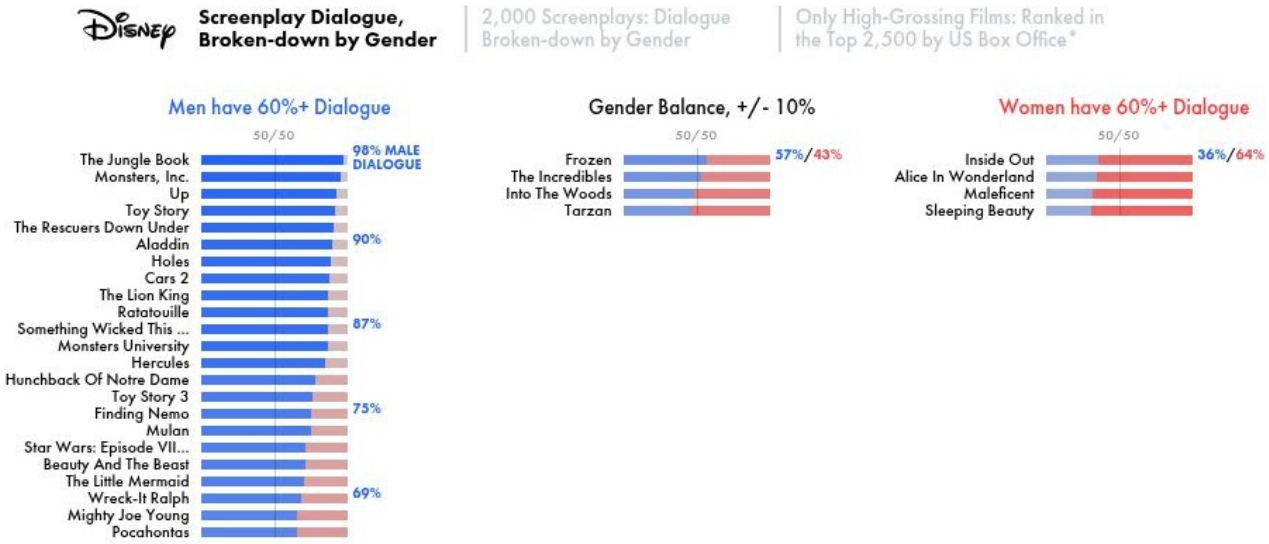
Capabilities to investigate. Group-relative lookup of baseline values, preserving nulls and different series endpoints, explicit denominator/baseline scope, and chart/table controls that consistently name their affected data.

Graphical edits that should update text. Selected measures and countries, baseline year, date range, colors, percentage-axis formatting, annotations and line-end labels should update the corresponding text definition.

Control scope. Country/date selections must state whether they apply to one output or a shared group of blocks. In the inspected source, switching the UK chart to Table initially exposed all 242 entities; a table search was needed to isolate the UK. That observed scope mismatch is informative.

Execution and editing lesson. Baseline transforms and alternate metric definitions should be inspectable beside the chart. A country change can invalidate a country-specific prose conclusion, so execution/state needs to distinguish fixed editorial examples from reader-selected results. The central calculation does not require a general statistical runtime.

Let’s begin by examining dialogue, by gender, for just Disney films.



In January 2016, researchers reported that men speak more often than women in Disney’s princess films. We validated this claim and doubled the sample size to 30 Disney films, including Pixar. The results: 22 of 30 Disney films have a male majority of dialogue. Even films with female leads, such as Mulan, the dialogue swings male. Mushu, her protector dragon, has 50% more words of dialogue than Mulan herself.

This dataset isn’t perfect. As with Mulan, a plot can center around a character, even though the dialogue doesn’t reflect it. And all of our data is based on screenplays, not a perfect transcription of a film.

11. Film Dialogue from 2,000 screenplays, Broken Down by Gender and Age - Screenshot 1. The Disney comparison: gender word-share bars followed by interpretation and the qualification that screenplays do not exactly transcribe the completed films. Direct source. Geometric excerpt of the captured page; the uncropped capture is included in the ZIP.

Data journalism / The Pudding / Grouped shares > within-film ranks > age comparison > drilldown

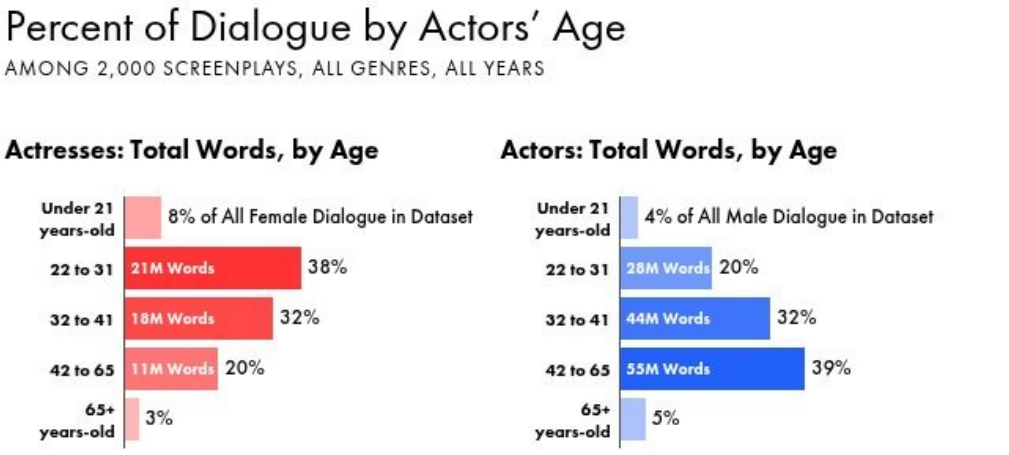
What the analysis does

The essay moves from a familiar subset (Disney) to all films, asks whether having a female lead changes total dialogue balance, then examines how words are distributed across actors' ages. The distinction between a lead character and all supporting dialogue is the analytical turn: a story can center a woman while most words still go to male characters. Prose also exposes the measurement limits before asking readers to explore individual films. This is a visual essay with calculations described in prose rather than visible code cells. The first graphic has three groups and the age graphic two facets, but each is one sequential analytical block; the optional final explorer is more dashboard-like.

Role of prose. Prose starts with a familiar subset, introduces measurement limits immediately after the first result, distinguishes lead roles from the contribution of the whole cast, motivates the age question, and then invites individual-film inspection.

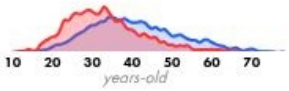
Aging out of Hollywood: Men vs. Women

For each film, we also determined the age of each cast member at the time of its release. This allowed us to quantify whether there is a bias toward younger women in Hollywood (or conversely, whether men enjoy a longer career).



The amount of dialogue, by age-range, is completely opposite for women versus men. Dialogue available to women who are over 40 years old decrease substantially. For men, it's the exact opposite: there are *more roles* available to older actors.

Here's another look at the same data, but for every age:

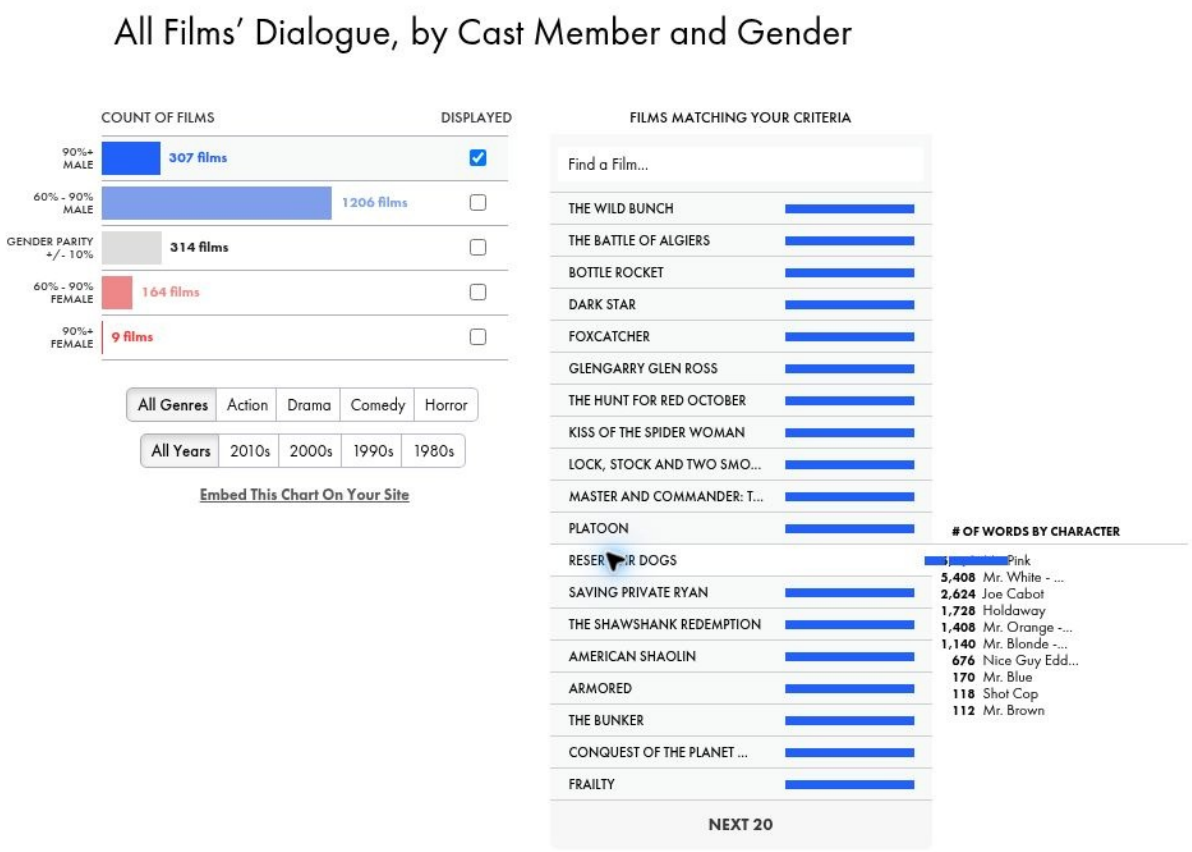


Why we made this

This project was born out of the less-than-stellar response to our analysis of films that fail the Bechdel Test. Commenters were quick to point out that the Bechdel Test is flawed and there are *justifiable* reasons for films to fail (e.g., they are historic). By measuring dialogue, we have much more objective view of gender in film.

11. Film Dialogue from 2,000 screenplays, Broken Down by Gender and Age - Screenshot 2. A later analytical step: age at film release is explained before a word-weighted age comparison and its interpretation. Direct source

addresses concerns about the methodology and data. Or if you don't know how to code, here's an easy way to comb through every film, genre, and year.



11. Film Dialogue from 2,000 screenplays, Broken Down by Gender and Age - Screenshot 3. Optional interaction evidence: selecting Reservoir Dogs changes the adjacent character-word breakdown. The explorer is more dashboard-like than the main essay and need not define the notebook layout. Direct source

Source data and calculations

Original source. Original material is screenplays found online, character matching to IMDb actor/actress entries, and actor age at film release. The public source-matching spreadsheet is linked by the article but not inspected. Original copyrighted screenplay texts are not provided as a complete downloadable corpus.

Row grain and scope. Character grain: one matched screenplay character; 23,048 rows, fields script_id, imdb_character_name, words, gender, age. Age is the actor's age, not the character's. Words, not lines, is the central count despite the HTML page title mentioning lines. Film grain: one screenplay/film, 2,000 rows, script_id, imdb_id, title, year, gross, lines_data. Actual CSV film years span 1929-2015. Join on script_id. The public core CSV does not include genre or a Disney flag. Domestic gross is described as inflation-adjusted, but unit/base year was not verified.

Fields and units. Character data: script_id, imdb_character_name, words, gender, age (actor age at release). Film data: script_id, imdb_id, title, year, gross, lines_data. Gender is a binary coding with five unknown rows. Join on script_id; genre and Disney membership are absent from these core files.

- Author repository

- character CSV

- film metadata CSV

- source-matching spreadsheet

Calculations, dependencies and stated limits

- Published filtering: characters with at least 100 words are matched; the article says at least 90% of a script's lines are assigned by gender. Minimum word count in released character CSV is 101. Small roles and script-to-screen changes affect individual-film results.

- Necessary transformations are transparent conceptually: sum words by film and gender, divide by film total for share; rank characters by words within film; count female lead/second/top-three roles; group actor ages and sum words. Our independent check of released rows gives 436 female leading-word characters out of 2,000 films (21.8%), consistent with the article's rounded 22%.

What feeds later results. Character rows + film metadata > joined table > grouped word totals/share denominator and within-film ranks. Those feed film summaries. Cleaned actor ages feed a separate word-weighted age view. Selected script_id filters the character detail.

Reproduction sufficiency. The data suffice for film shares, character ranks and a documented exploratory age variant. They do not reproduce the initial extraction, exact genre filters, or the original age calculation without further source work. Suitable substitute: use the released aggregate CSVs, select Disney films by an explicitly listed title set, omit genre/gross filters, report cleaned-age coverage.

Limits and uncertainty. Exact original chart code, tie policy, unknown-gender denominator and age-cleaning rules were not verified. Released data has five "?" gender rows, 4,785 NULL ages and 16 implausible ages above 110 (some values resemble release years). These need an explicit cleaning rule; do not silently reproduce the age percentages as exact. Displayed age bins are "Under 21," "22 to 31," "32 to 41," "42 to 65," "65+"; exact inclusive endpoints are ambiguous without code.

Elements needed to reproduce it

1. question and measurement definition
2. selected-film gender word sums
3. normalized bars
4. caveat/interpretation
5. all-film shares
6. lead-versus-total ranking
7. age-at-release grouping
8. age comparison
9. qualified conclusion
10. optional individual-film inspection

Element	Inputs	Calculation or configuration	Visible result	Interaction	Dependency
Measurement explanation	Methods/source coverage	Define word count, threshold and script limitations	Prose	Source links observed	None
Film shares	Character CSV + film titles	Sum words by film/gender; normalized share; sort	100% horizontal bars; color=gender; 50% reference	Subset tabs observed; exact Disney membership needs definition	Join and group totals
Interpretation	Selected films and totals	Contrast lead character with total cast dialogue	Prose and highlighted examples	None	Film shares
Lead/rank summary	Character words	Descending rank within film; counts by rank/gender	Percentages and comparison	Proposed rank selection	Character rows

Element	Inputs	Calculation or configuration	Visible result	Interaction	Dependency
Age comparison	Valid actor ages, words, gender	Explicit age bins; word-weighted sum and within-gender denominator	Grouped/faceted bars, percent labels	Static observed	Cleaned age rows
Explorer controls	Film shares and metadata	Gender-share band, decade, genre, title	Matching film list with bars	Share checkboxes, years/genres, search observed	Film-level table
Selected-film detail	Selected script_id	Filter characters; sort words descending	Character counts and bars	Reservoir Dogs selection tested successfully	Explorer selection
Scope/caveats	Missingness and methods	State exclusions and limits	Prose/coverage table	Optional table inspection	Preparation
Essential versus optional	Reference scope	documented aggregation/denominator, film share comparison, rank contrast, method caveats. Optional: age section until cleaning is settled; custom Disney typography; all-film animated transitions; narrow tooltip layout; exact genre/gross filters; approximate dialogue timeline.	Faithful reduced version versus polish	As described above	Do not require omitted custom features for the core

Fit for explorEDA

Smallest faithful version. A named film subset with normalized gender shares > all-film share summary > lead/rank comparison > film selection with character contributions. Conventional stacked bars and tables preserve the argument.

Likely fit, subject to verification. Tables, interactive charts, filters and calculated fields appear suitable for the aggregate-data core. Custom Disney styling and film graphics are unnecessary.

Capabilities to investigate. Joining character and film records, ranking/top-k inside groups, linking film selection to its character rows, and keeping percentage denominators explicit when filters change. Age reproduction also requires an explicit cleaning/coverage policy.

Graphical edits that should update text. Grouping fields, normalization, sort order, gender colors, the 50% reference, selected films, filter predicates and labels should update defining text.

Control scope. A selected film drives its character detail. A broader subset changes shares/rank summaries only in explicitly connected blocks. In the source, narrative summaries stay fixed while the optional explorer changes. Genre controls are observed, but the released core CSV lacks genre.

Execution and editing lesson. A visual filter can accidentally alter the denominator of a percentage. This reference tests aggregate-to-detail lineage, within-group ranking and the difference between fixed narrative examples and scoped exploratory selections.

Comparison

Example	Analytical pattern	Data availability	Calculation transparency	Interaction complexity	Effort
Plot Exploration: Penguins	Raw observations > grouped comparison > faceted distributions > summary overlay	Exact attachment and canonical public CSV; live rows verified	High: visible solution code; final requested count not supplied	Low: hover, table inspection and disclosure; editing is central	Low for three progressive scatters; medium with facets, bins and overlays
What's different? Analyzing Time-Series Forecast Performance	Actual versus forecast > signed difference > event-window investigation	Exact attachment; live 1,417 rows	High; explicit subtraction and fixed date filter	Low; authored views, no main reader controls	Low
What caused this? An analysis adventure	Aggregate anomaly > category decomposition > shared-scale comparisons	Exact cached JSON; live 9,912 rows	High for data shaping and chart aggregation; helper polish partly uninspected	Low in reading mode	Low-medium
Discovering Date Patterns	Time series > calendar features > grouped distributions	Exact attachment; live rows verified	High; explicit derivation and grouping; window defaults unresolved	Moderate: brush, rolling and facet selectors	Medium
Correlation over time	Time-aligned join > visual relationship > cal endar-conditioned comparisons	Exact two-file attachments; live rows and matching archived weather bytes	High for join and marks; precomputed category cutoffs partly opaque	Moderate: two radios and shared palette	Medium
How Bratenahl, Ohio, Disproportionately Targets Black Drivers	Missing-data sensitivity > grouped rates > association > geographic corroboration	Live CSVs and geometry attachment; original Census vintage unresolved	High: source formulas visible; one broken branch and denominator inconsistencies explicit	Low-moderate: hover plus one scenario slider; no elaborate graphics	Medium for officer section; high for full sensi tivity/spatial investigation
Local Spatial Autocorrelation Interactive Tutorial	Neighbor definition > standardization > spatial lag > global/local inference	Exact GeoJSON and original ZIP/codebook; live 85 records	High-level formulas and source calls visible; library internals and random seed not fully resolved	High; linked selection and stochastic controls; reducible	High
Is Seattle Really Seeing an Uptick In Cycling?	Time series > confounder adjustment > residual trend	Exact author CSVs and notebook accessible; blank counter cells need a declared policy	High: full code; covariance errors and window prose/code mismatch require review	Low in source: static outputs; m odel/date/windo w controls would be proposed	High for adjustment chain; medium for summaries alone
Get started with tidymodels and #TidyTuesday Palmer penguins	Explore > fit and validate > revise the visual encoding	Tidy CSV fetched; original species data links documented	High: inline R code, partitions, model outputs and coefficients	Low in source: static analytical walkthrough	Medium with one fit; high with two models and resampling
Many countries have decoupled economic growth from CO_2 emissions, even if we take offshored production into account	Normalize time series > compare definitions > compare countries	Exact chart CSV, metadata and README downloaded; 8,622 entity-years	Medium/high: prepared metrics documented and normalization checked; missing-baseline behavior unresolved	Moderate: country, timeline, chart/table and download controls	Low/medium

Example	Analytical pattern	Data availability	Calculation transparency	Interaction complexity	Effort
Film Dialogue from 2,000 screenplays, Broken Down by Gender and Age	Grouped shares > within-film ranks > age comparison > drilldown	Character and film aggregate CSVs fetched; scripts/genre metadata and age-cleaning rules incomplete	Medium: methods described, core aggregates available, edge rules hidden	Moderate: subset/filter/search and linked film selection, tested	Medium for share/rank core; high for exact full replica

Low means prepared data with standard summaries or row calculations. Medium adds joins, windows, facets, normalization or scoped controls. High adds model fitting, resampling, spatial operations or reconstruction of partly hidden preparation. These ratings assume the smallest faithful version, with optional polish omitted.

Three recommended starting examples

SMALL FIRST PROOF

What's different? Timeseries forecast performance

Faithful scope. Keep a source table, actual-versus-forecast line chart, a named delta = actuals - forecast calculation, a signed difference chart, one event annotation and the interpretation connecting them.

Distinct capability tested. This isolates the central promise: a small textual calculation feeds multiple full-width outputs, while graphical changes to encodings, the zero line, colors and annotation persist to the appropriate block's text.

The dataset is already prepared and the arithmetic is transparent. A useful first proof can be small without losing the argument. Resolve the source's inconsistent power/energy labels before presenting a polished reproduction.

BROADER REPRESENTATIVE WORKFLOW

Discovering Date Patterns

Faithful scope. Preserve raw-data inspection, a rolling time series, local calendar-field derivation, a derived table and a selectable grouped comparison.

Distinct capability tested. This exercises reusable intermediate data, ordered windows, timezone semantics and multiple control scopes. A facet edit changes the grouping configuration and dependent group means; a local brush should retain its declared scope.

It combines the kinds of tables, calculations, filters and graphical editing already described for explorEDA. The two local calculation branches make it a stronger workflow test than simply placing several charts beneath paragraphs.

STRETCH CASE

Is Seattle Really Seeing an Uptick In Cycling?

Faithful scope. Preserve the count summaries, weather/date join, one explicitly specified adjustment model, predictions and corrected series, with a qualified interpretation.

Distinct capability tested. This tests joins, model-result tables, dependent calculations and selective re-execution. Cosmetic chart edits must remain distinct from changing a predictor, a training/data scope or a derived quantity.

The analytical argument depends on each preceding calculation, while the graphics remain conventional. Reproduce the reasoning structure first; review the legacy model specification and uncertainty calculation before treating its historical numerical estimate as validated.

Recurring notebook patterns worth supporting

A full-width sequence of distinct block types. Narrative, named calculation/query, inspectable table, chart and interpretation should coexist in one reading column. A block needs a stable identity beyond its current position.

Reusable intermediate results. Derived columns, grouped tables, joined datasets and model outputs feed later blocks. Reordering the essay should not silently redefine calculation dependencies.

Explicit control scope. A date brush can change only a view, a filter can change several downstream results, and a selector can redefine grouping. Those different effects should be visible in the defining text and saved state.

Semantic graphical edits. Round-trip field choices, aggregation, bins/windows, facets, category order, scales, colors, legends, tooltips and annotations. Preserve unrelated calculation text and stable references while doing so.

Inspectable data contracts. Expose row grain, units, geography, date range, null policy, join keys and normalization baseline. Counts, shares, per-capita measures and indexes should not become interchangeable because their charts look similar.

Narrative validity after edits. A data-dependent numerical statement can refer to a result; a human interpretation may require review after an upstream change. Graphical editing should not leave the notebook appearing internally consistent when the prose describes an earlier state.

Execution state and reproducibility. Distinguish cheap reactive calculations from explicit Run operations. Preserve inputs and available random seed/state, show which blocks failed or are stale, and keep unaffected branches readable.

A chart and its supporting table. Switching between visual and tabular inspection helps explain filters, transformed values and missing endpoints. The views should retain shared data provenance even when presentation differs.

Unusual features specific to individual examples

- Penguin exploration: pedagogical Hint/Solution reveal controls are specific to a guided exercise; ordinary explanatory text and visible calculation blocks can carry the same analytic progression.

- Calendar and energy stories: hand-annotated screenshots and hand-drawn story illustrations are optional editorial polish. Their findings can be expressed with ordinary chart annotations and prose.

- Spatial tutorial: polygon contiguity, spatial lags, conditional permutation tests and coordinated maps are specialized analytical capabilities. The linked side-by-side map arrangement is not a general notebook-layout requirement.

- Seattle: a solar-daylight formula and a sequence of increasingly elaborate regressions are specific modeling details. One documented daylight input and one sound adjustment model retain the central reasoning.

- Film dialogue: custom shapes, film search and a screenplay-matching pipeline are particular to this story. Film shares, within-film ranks and a selected-film character table form the core. Age comparisons are an optional extension until their cleaning rules are resolved.

- Modeling essays: exact historical random samples and package versions matter for byte-for-byte result matching, but do not imply that a notebook view must support arbitrary language execution.